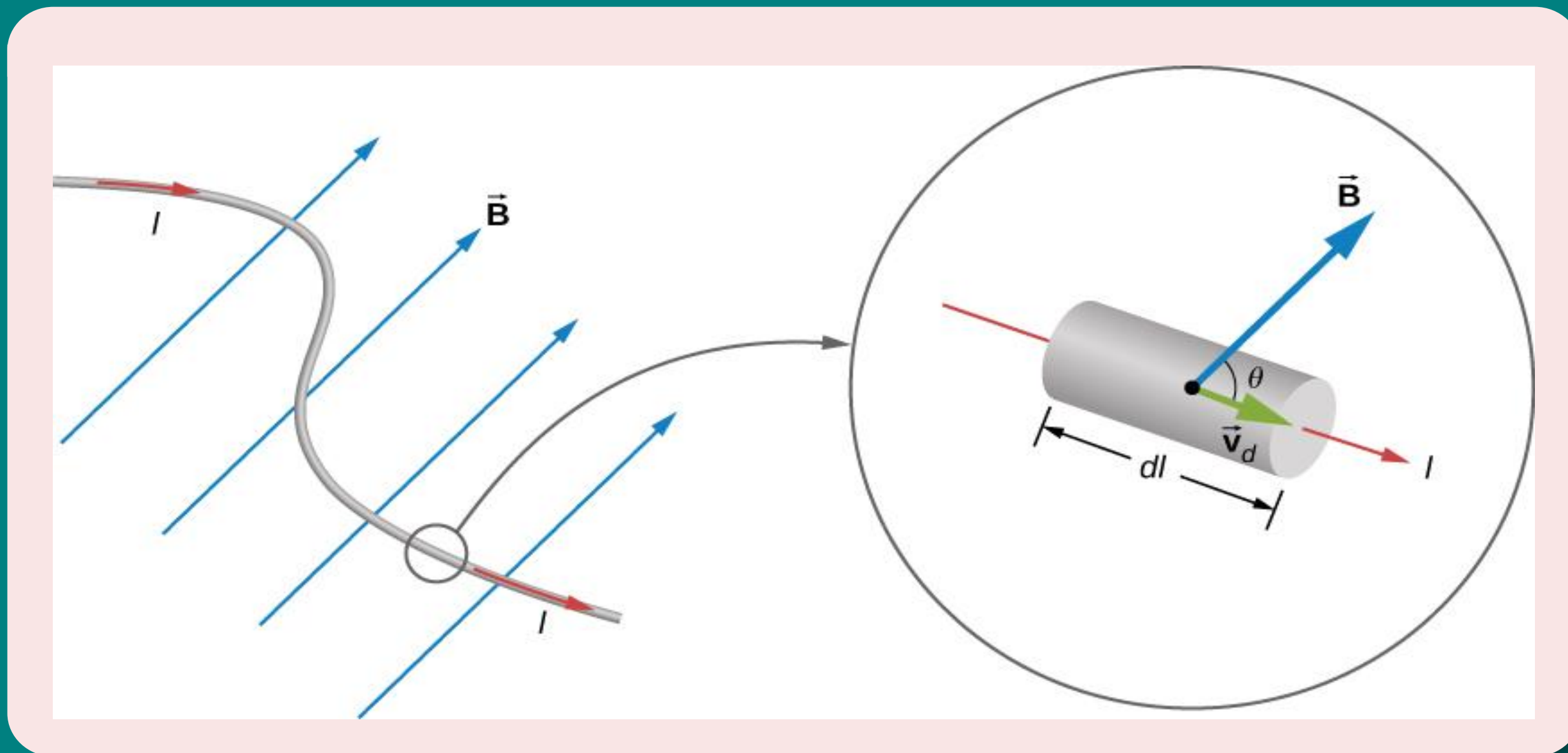
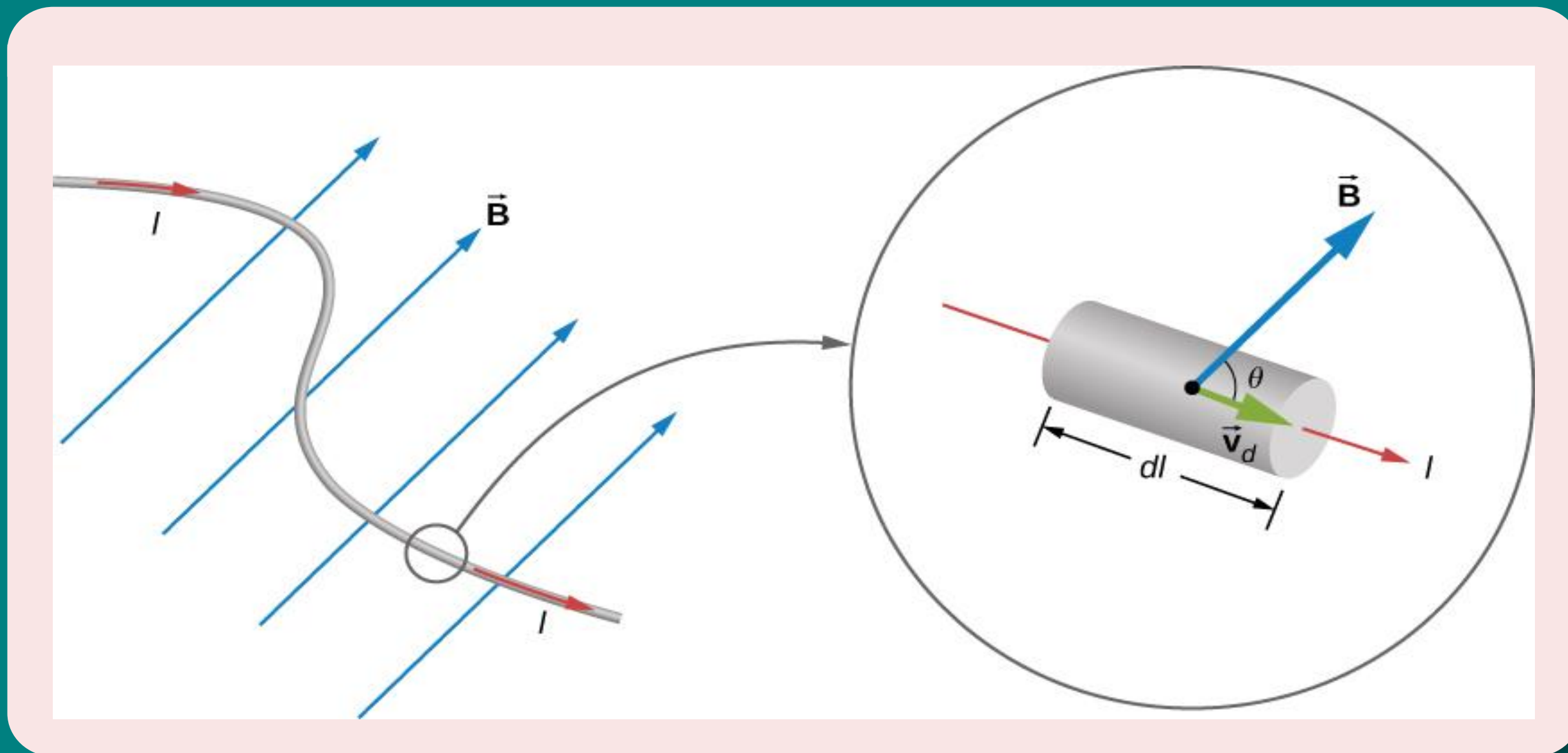


Magnetic force on a current-carrying conductor

- Electric current is **flow of charge**. A **current - carrying wire** in a **magnetic field** ($\vec{B}$) must therefore **experience a force** due to the field.

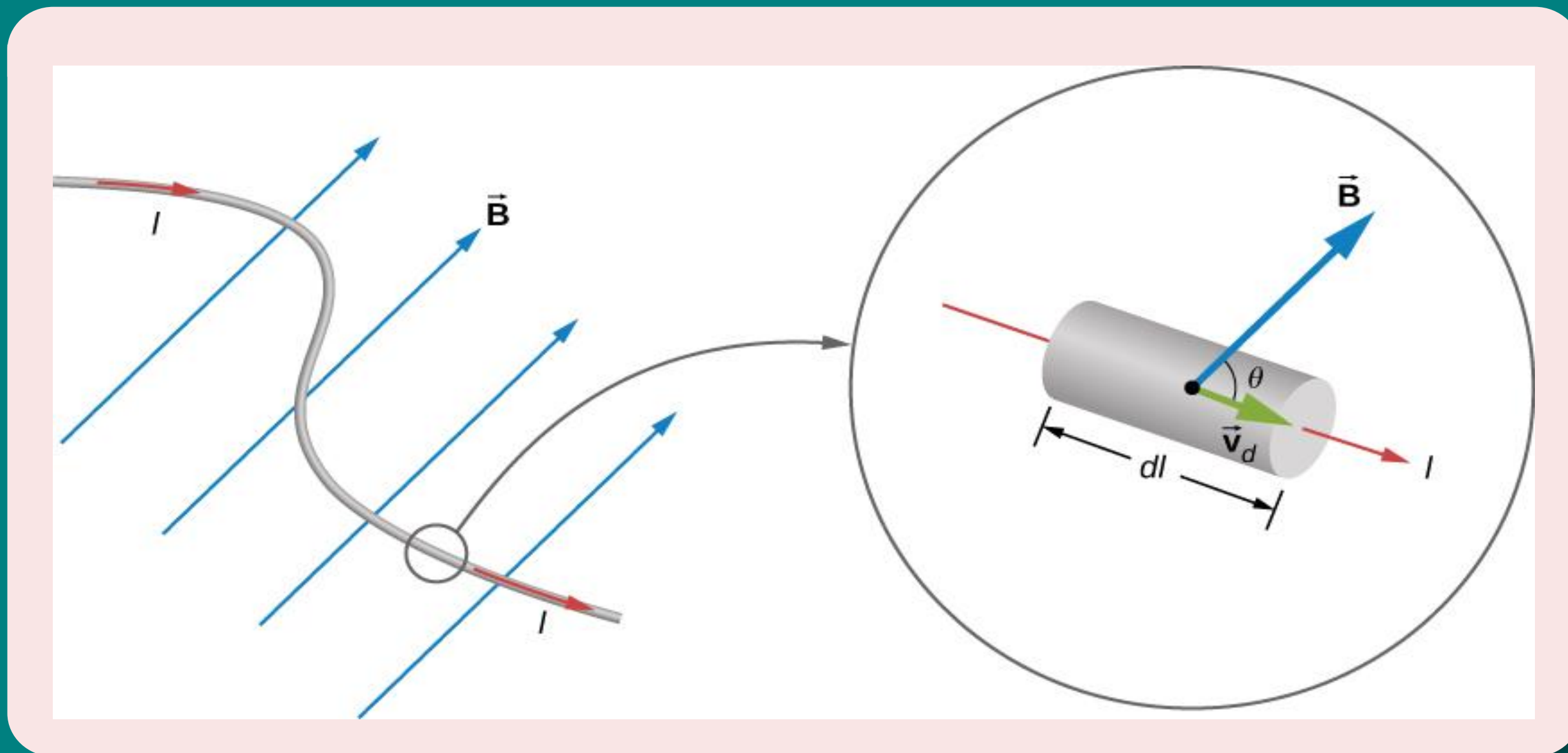


Magnetic force on a current-carrying conductor



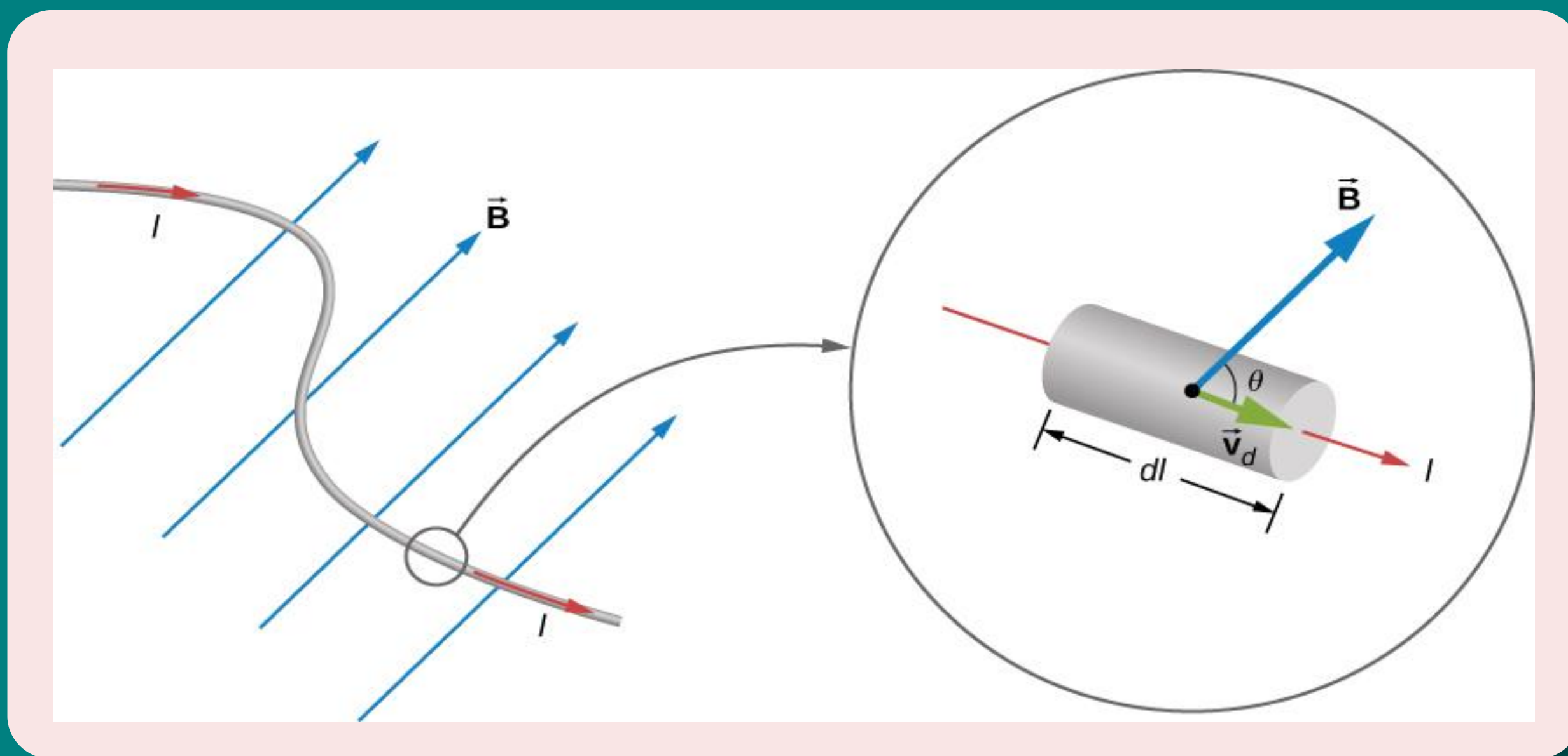
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Magnetic force on a current-carrying conductor



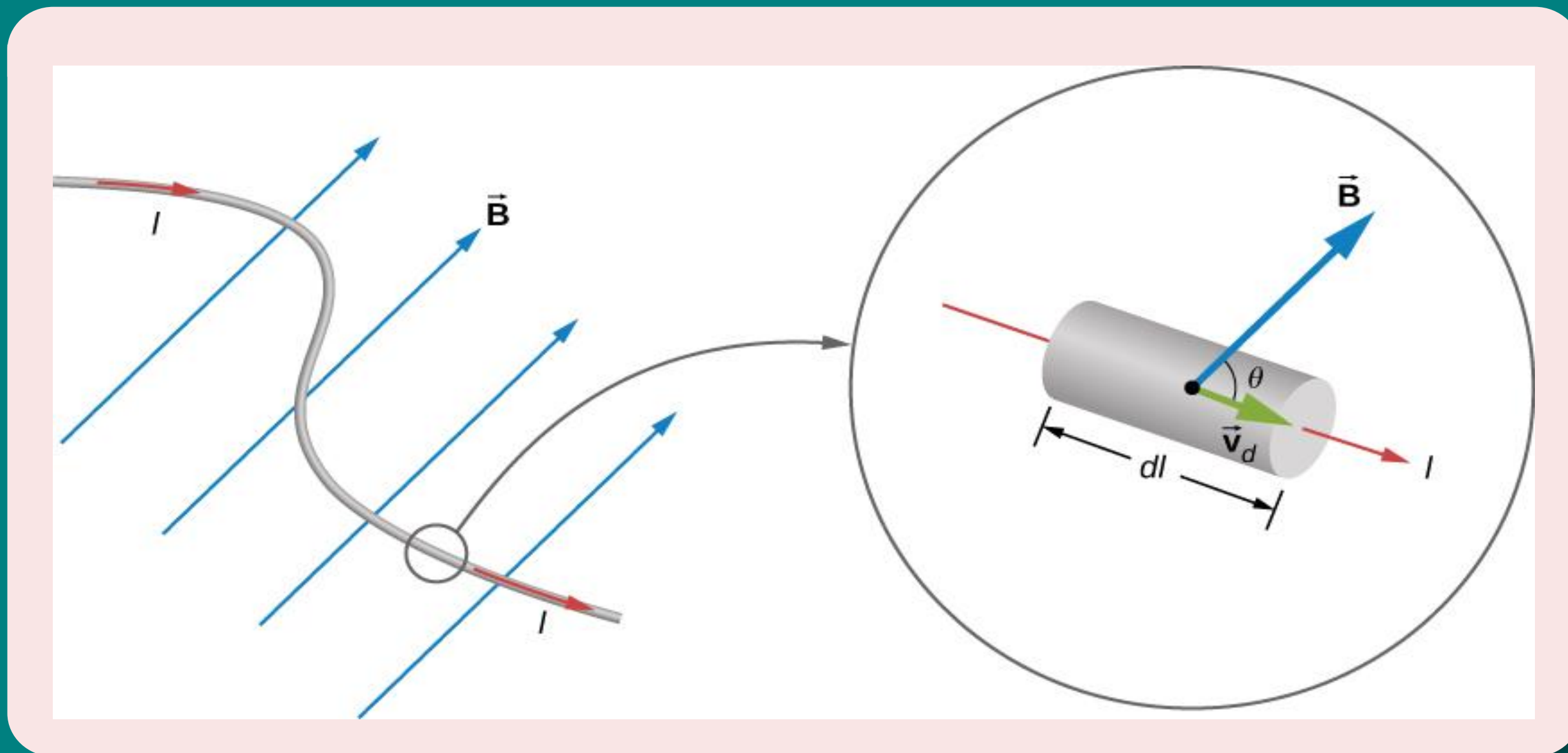
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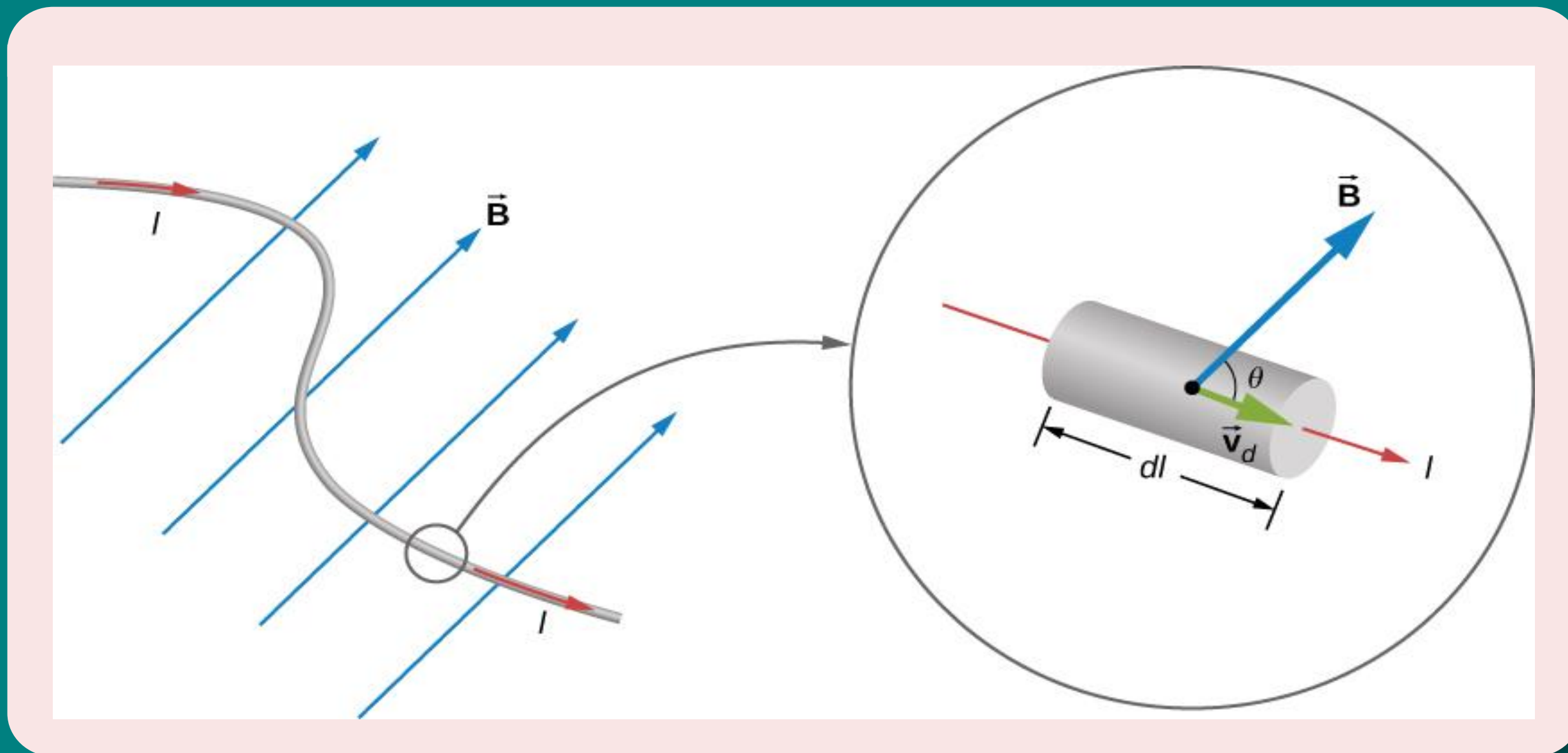
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Magnetic force on a current-carrying conductor



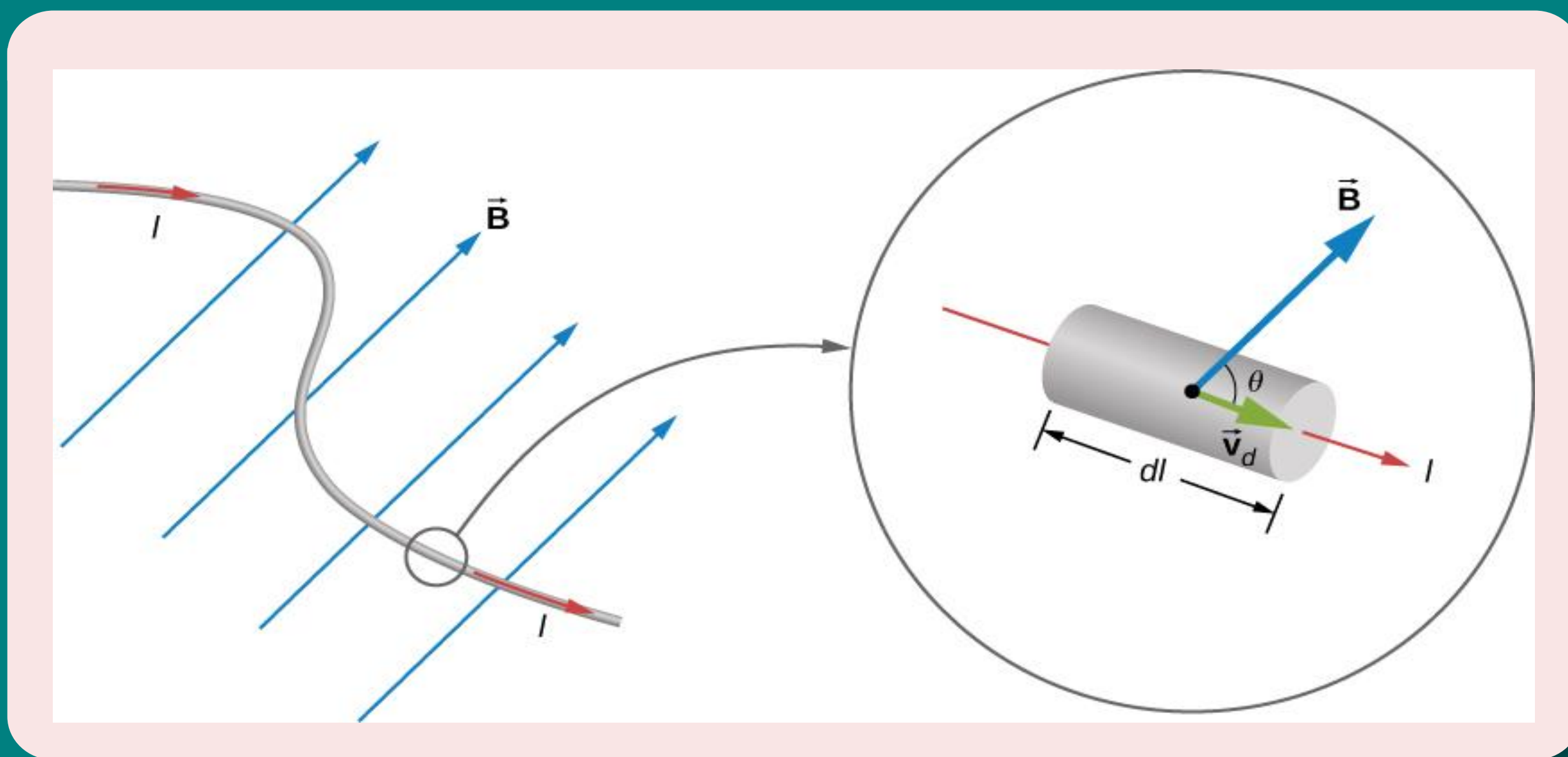
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Magnetic force on a current-carrying conductor



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- So its **volume** is $V = Adl$.
- Let, n be the **charge carriers per unit volume**.
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- If the **charge carriers** move with **drift velocity** $\vec{v}_d$

Magnetic force on a current-carrying conductor

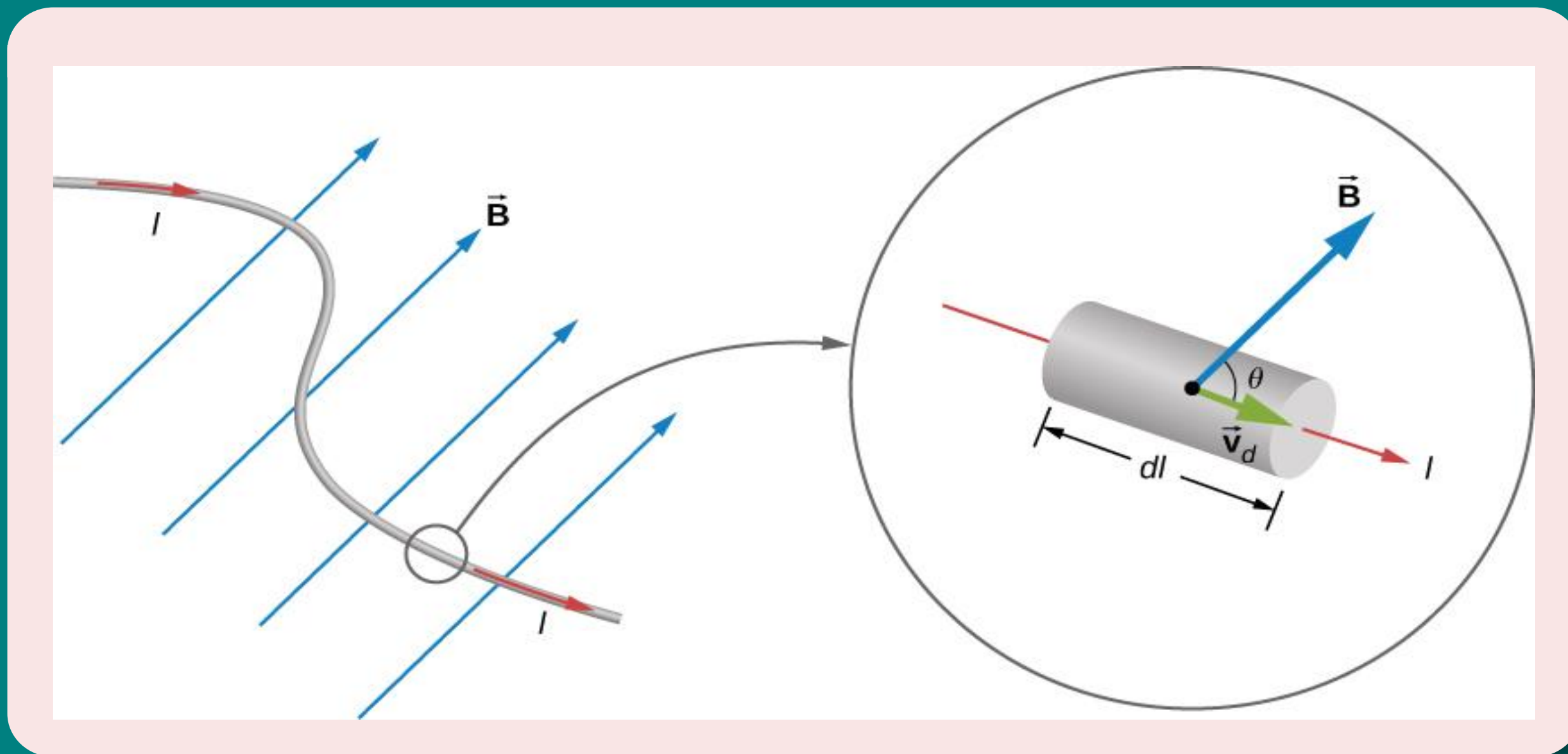


- **Force on the charge carriers** moving with velocity $\vec{v}_d$ in the section of wire is

$$d\vec{F} = Q\vec{v}_d \times \vec{B}$$
$$d\vec{F} = neAdl\vec{v}_d \times \vec{B}$$

- We can define $d\vec{l}$ to be a vector of length dl pointing along $\vec{v}_d$ (or I), which allows us to rewrite this equation as

Magnetic force on a current-carrying conductor



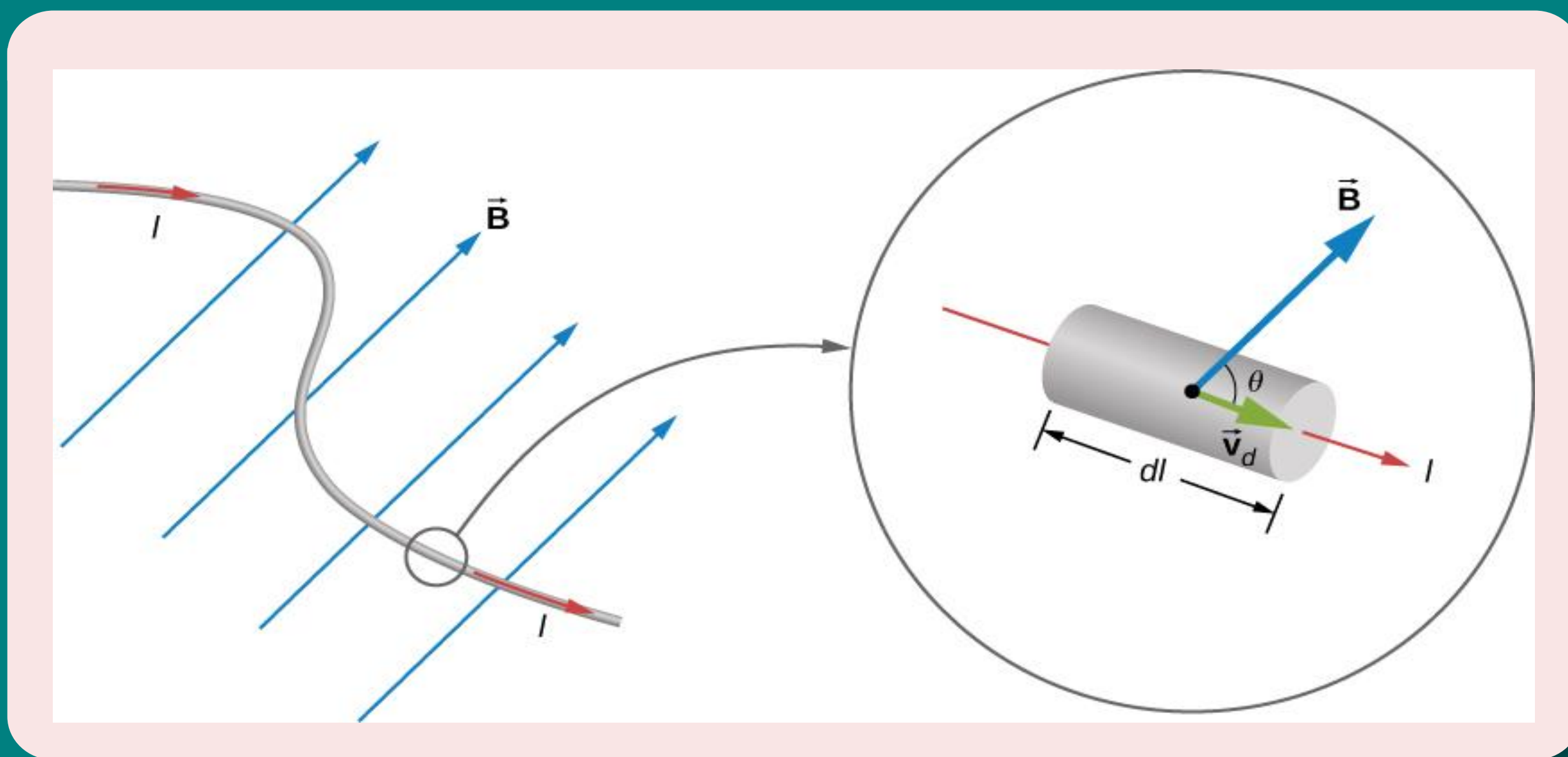
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$$d\vec{F} = neA v_d d\vec{l} \times \vec{B}$$

Magnetic force on a current-carrying conductor



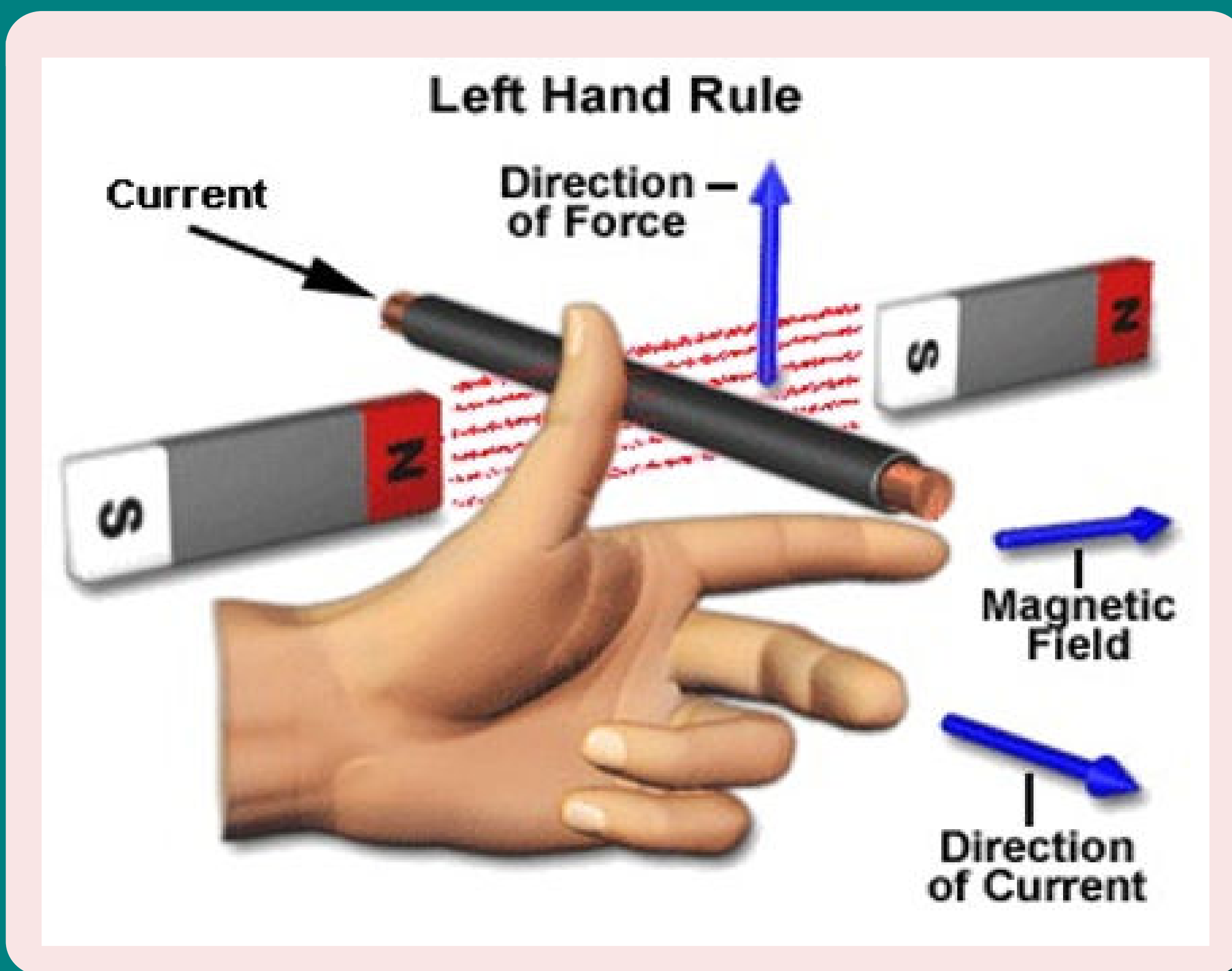
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- We can define $d\vec{l}$ to be a vector of length dl pointing along $\vec{v}_d$ (or I), which allows us to rewrite this equation as

$$d\vec{F} = neAv_d d\vec{l} \times \vec{B}$$
$$\boxed{d\vec{F} = I d\vec{l} \times \vec{B}} \quad (\because I = neAv_d)$$

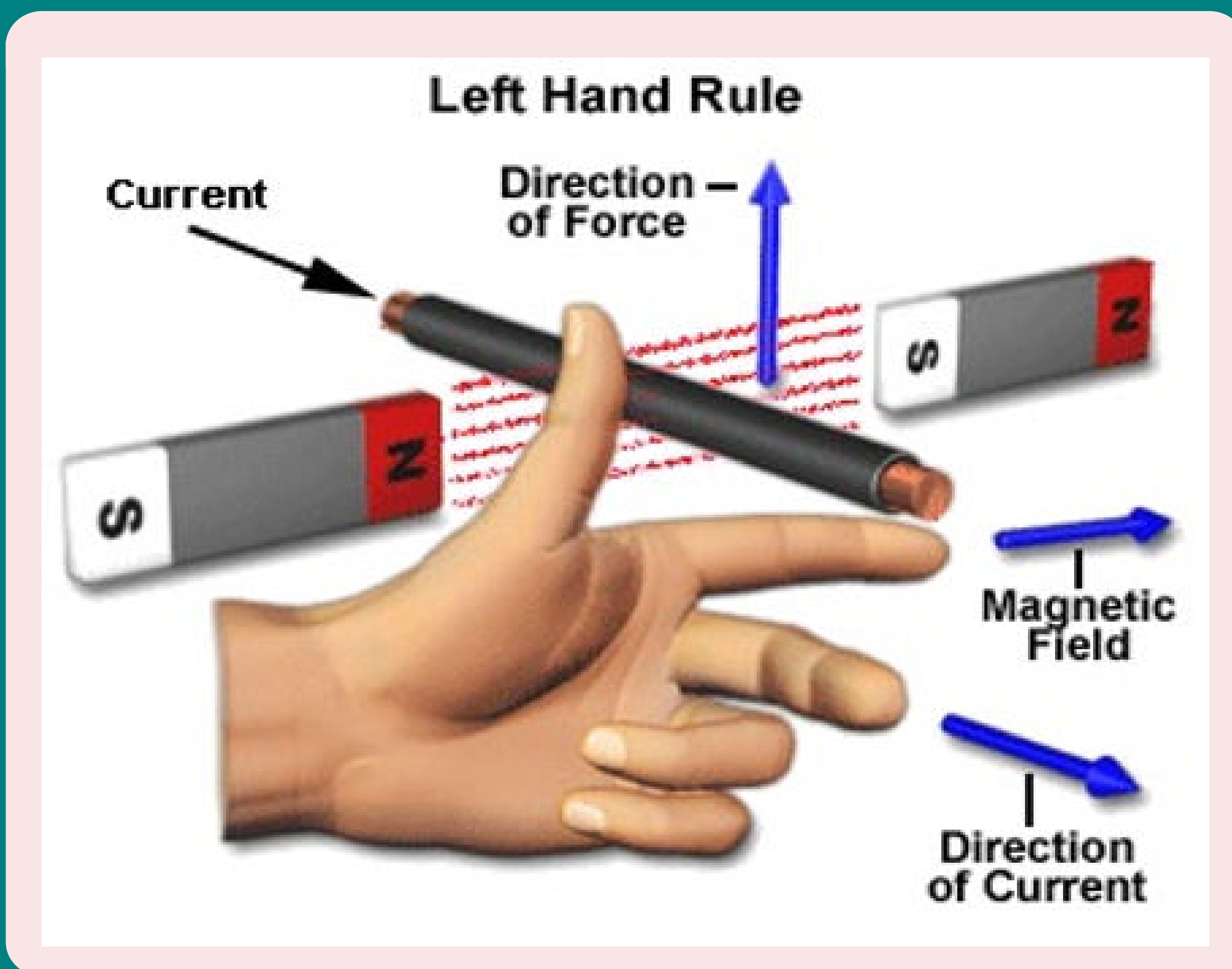
Magnetic force on a current-carrying conductor



- The force on a straight, current-carrying wire in a uniform magnetic field.

$$\vec{F} = I\vec{l} \times \vec{B}$$

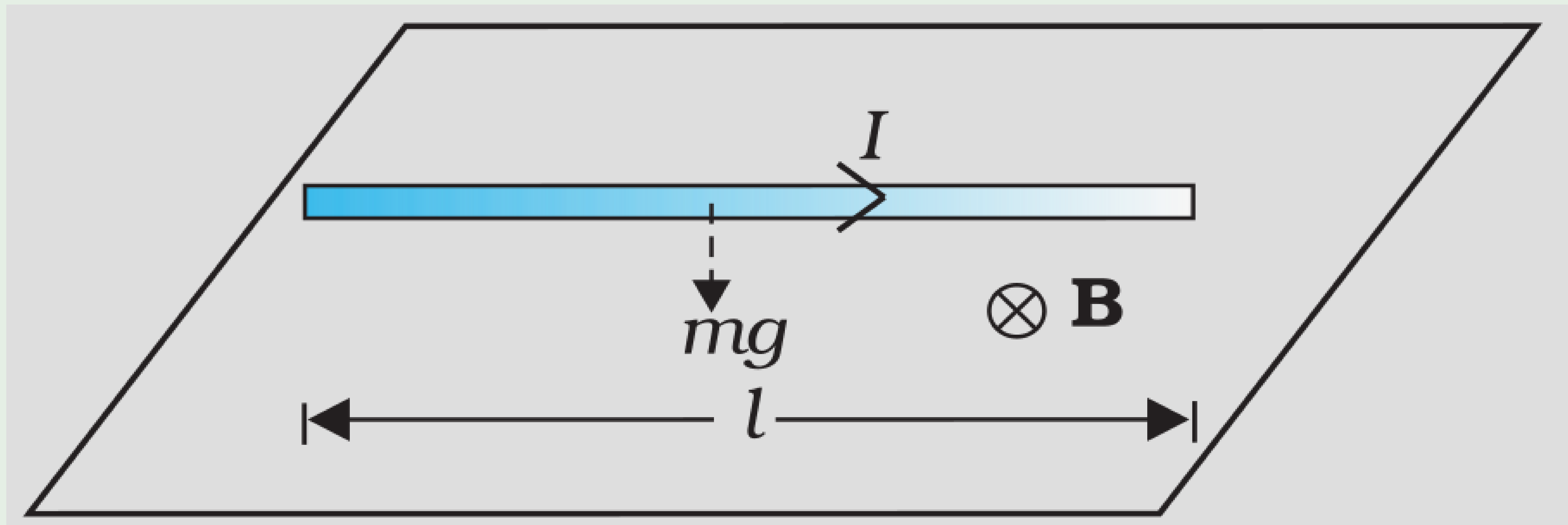
Magnetic force on a current-carrying conductor



- The force on a straight, current-carrying wire in a uniform magnetic field.
- $$\vec{F} = I\vec{l} \times \vec{B}$$
- The direction of this force is given by Fleming's Left hand rule.

Example 4.1

A straight wire of mass 200 g and length 1.5 m carries a current of 2 A . It is suspended in mid-air by a uniform horizontal magnetic field B . What is the magnitude of the magnetic field?



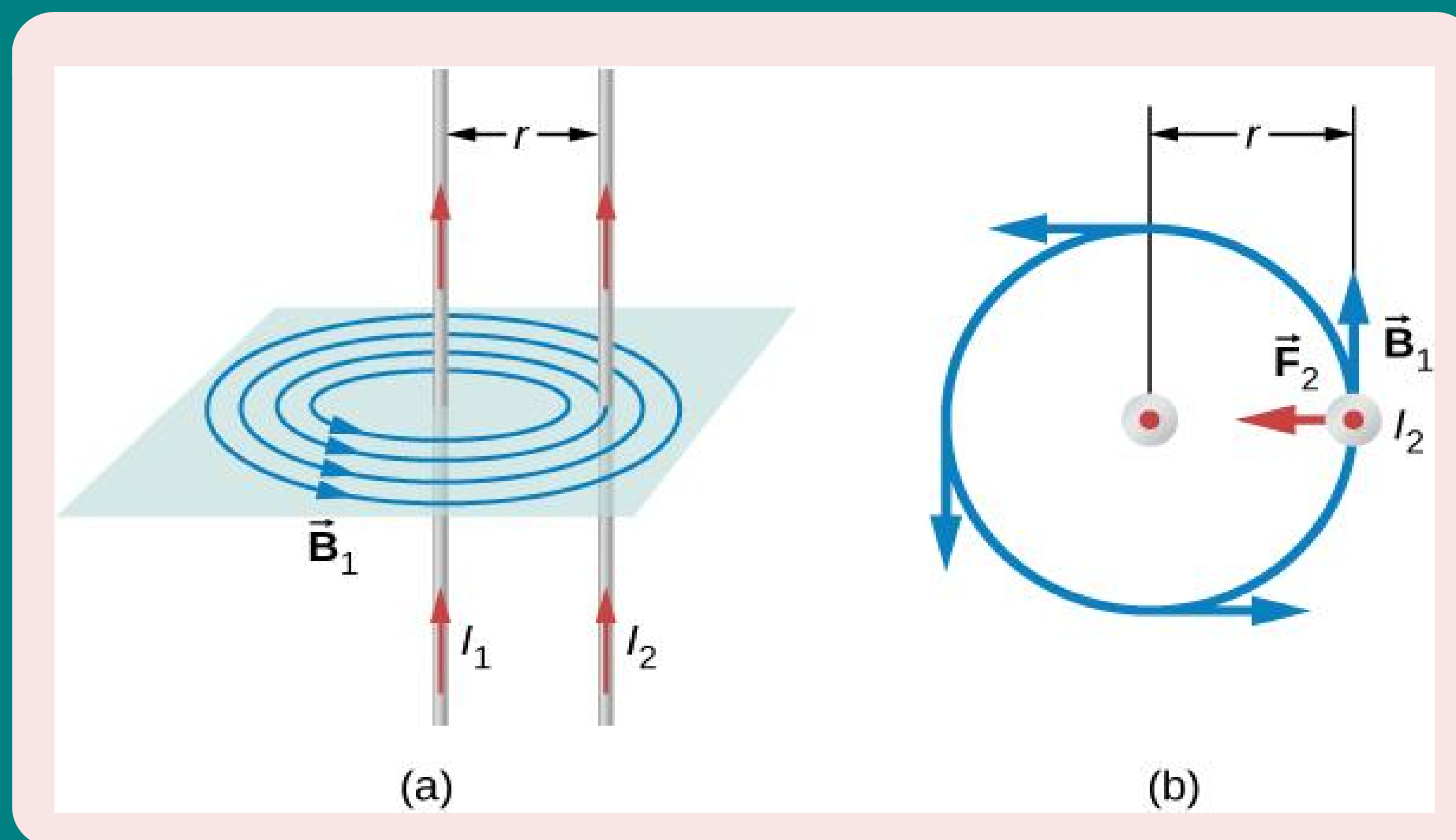
Exercise 4.5

What is the magnitude of magnetic force per unit length on a wire carrying a current of 8 A and making an angle of 30° with the direction of a uniform magnetic field of 0.15 T ?

Exercise 4.6

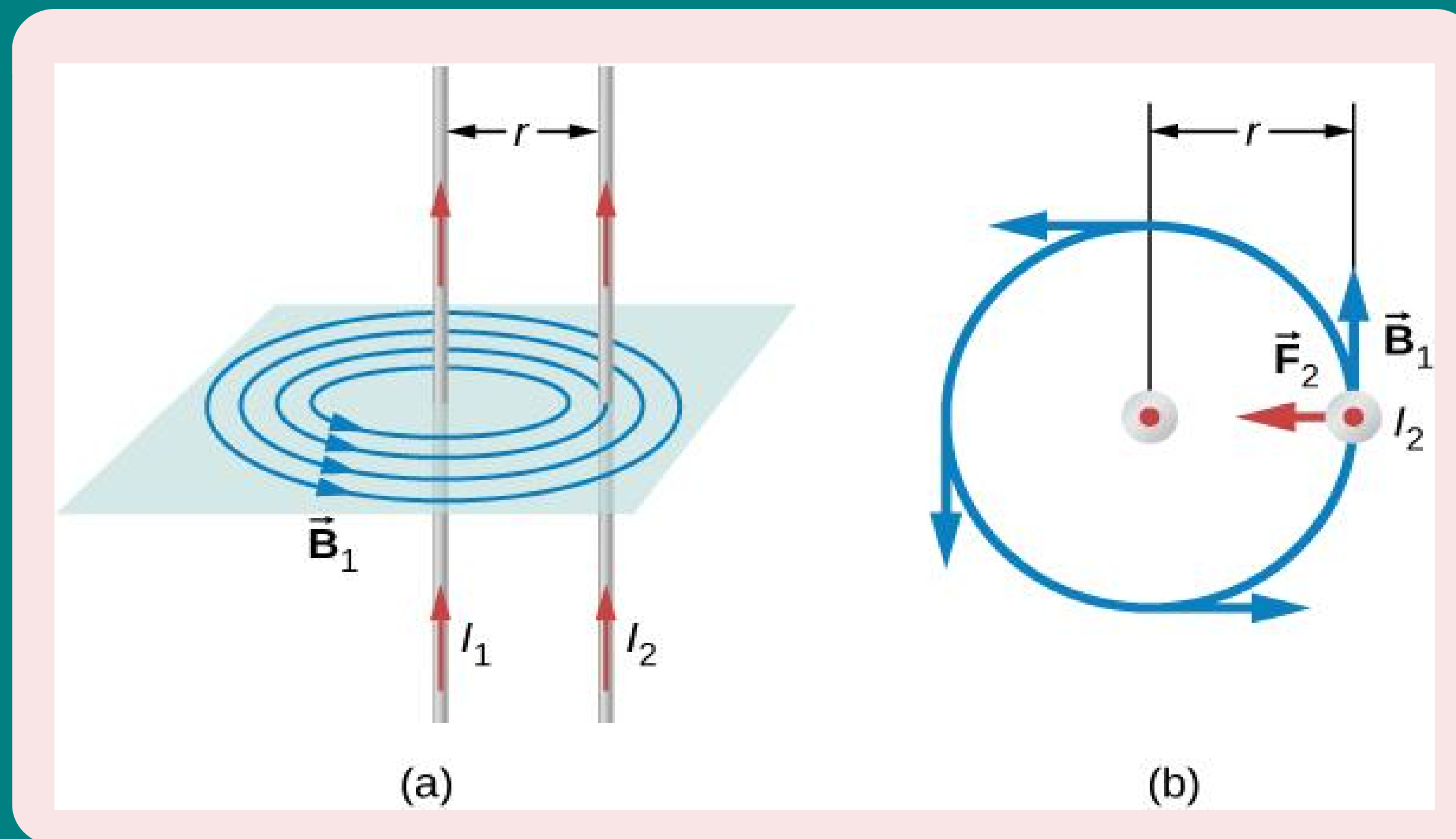
A 3.0 cm wire carrying a current of 10 A is placed inside a solenoid perpendicular to its axis. The magnetic field inside the solenoid is given to be 0.27 T. What is the magnetic force on the wire?

FORCE BETWEEN TWO PARALLEL CURRENTS



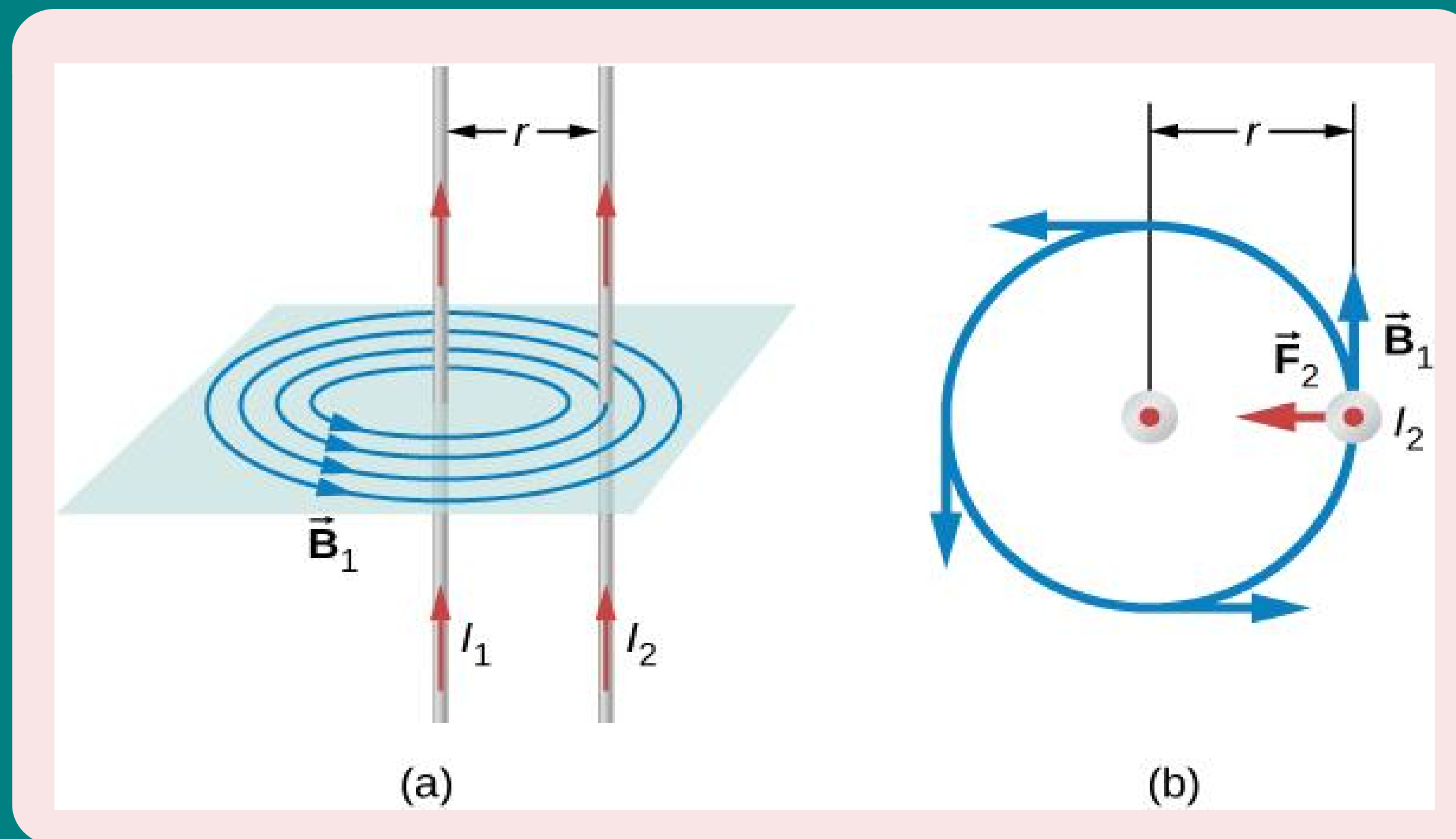
- We know that a **magnetic field is generated by a current-carrying conductor.**

FORCE BETWEEN TWO PARALLEL CURRENTS



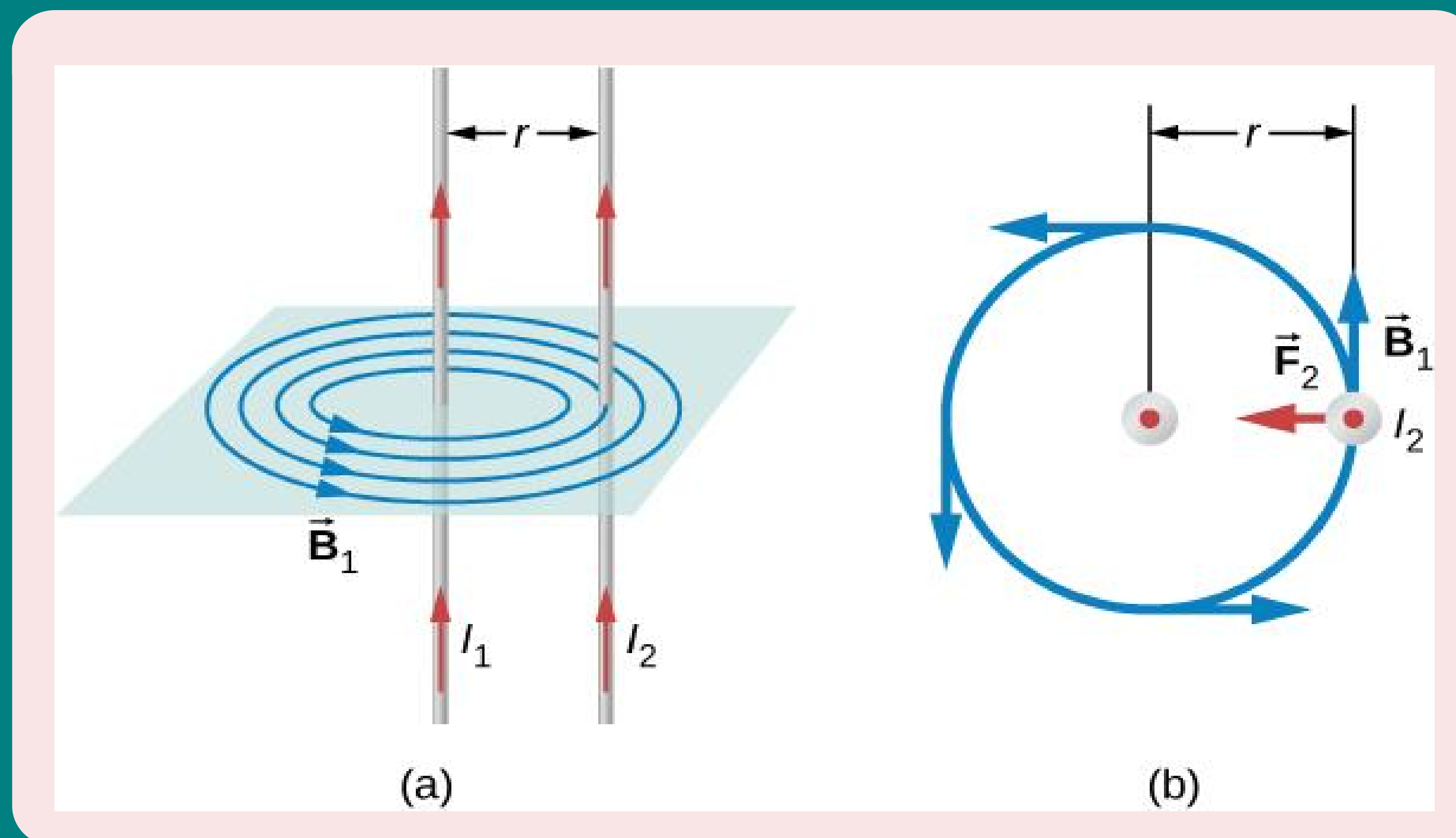
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- **Another current-carrying conductor experiences force as a result of the external magnetic field.**

FORCE BETWEEN TWO PARALLEL CURRENTS



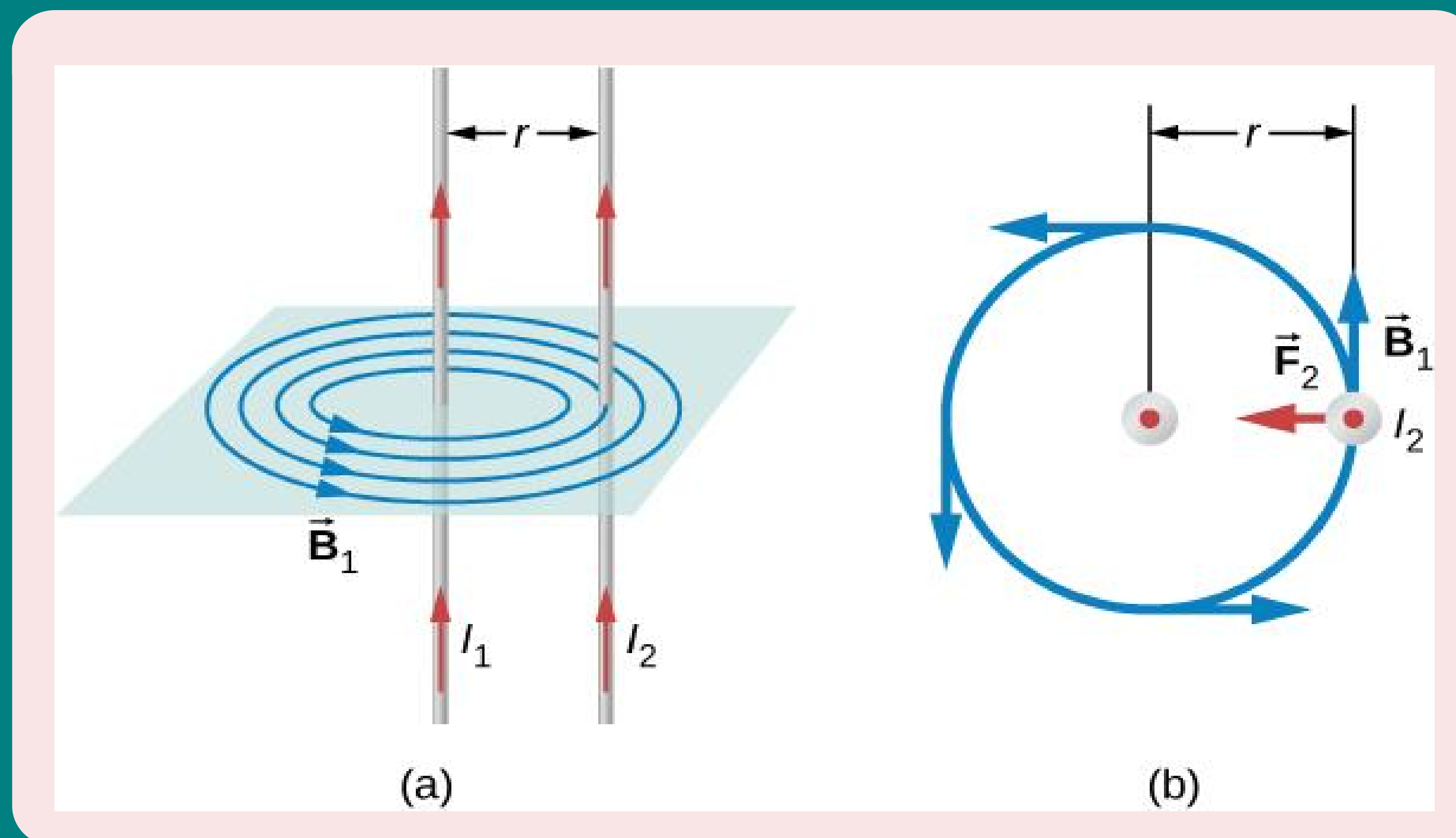
- We know that a **magnetic field** is generated by a **current-carrying conductor**.
- **Another current-carrying conductor experiences force** as a result of the **external magnetic field**.
- Consider **two long parallel current-carrying wires**, separated by a **distance 'r'**, such that **one of the wires is carrying current I_1** and the **other is carrying I_2** .

FORCE BETWEEN TWO PARALLEL CURRENTS



- The magnitude of the field due to I_1 at a distance r is

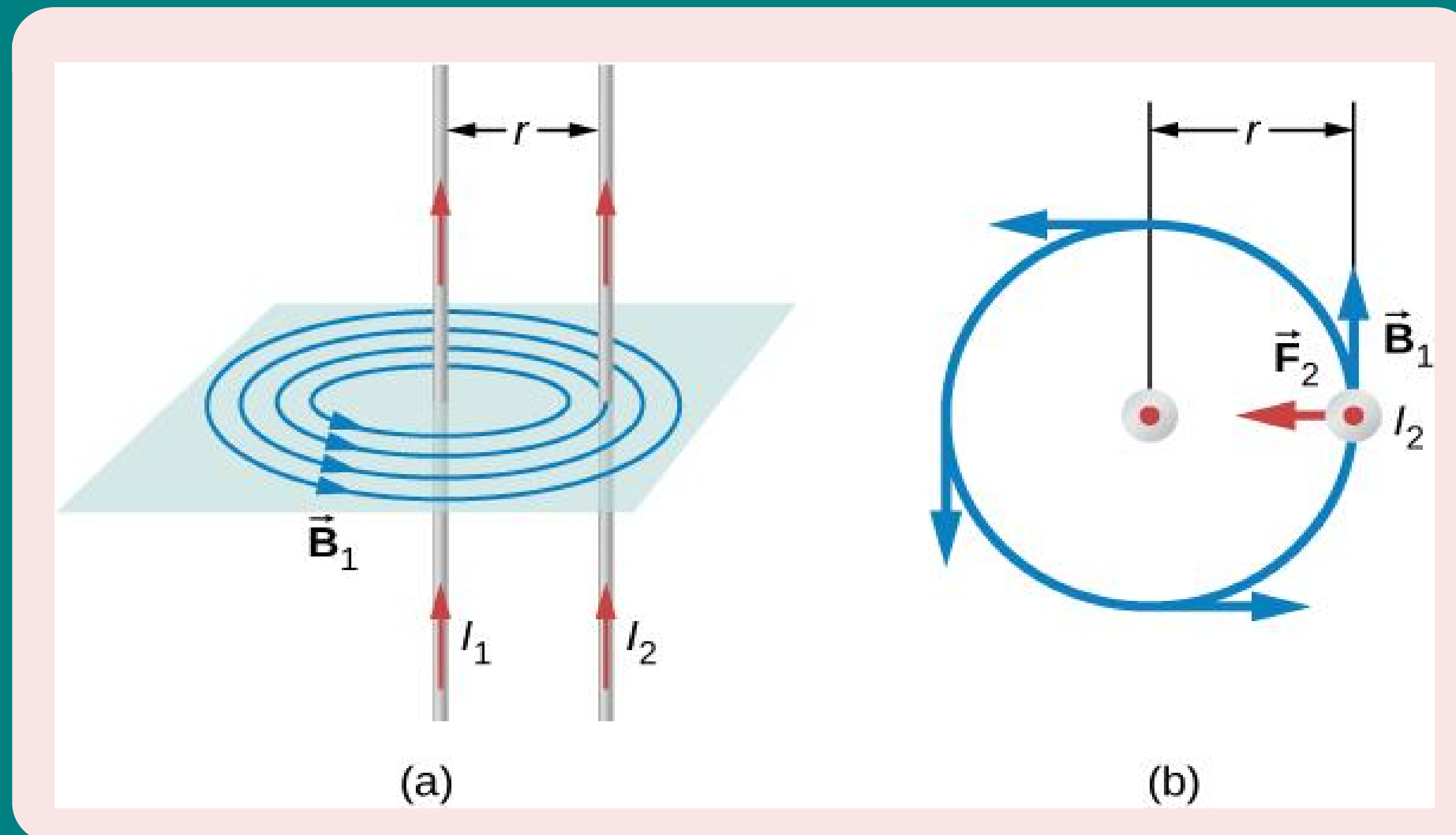
FORCE BETWEEN TWO PARALLEL CURRENTS



- The magnitude of the field due to I_1 at a distance r is

$$B_1 = \frac{\mu_0 I_1}{2\pi r} \dots (1)$$

FORCE BETWEEN TWO PARALLEL CURRENTS

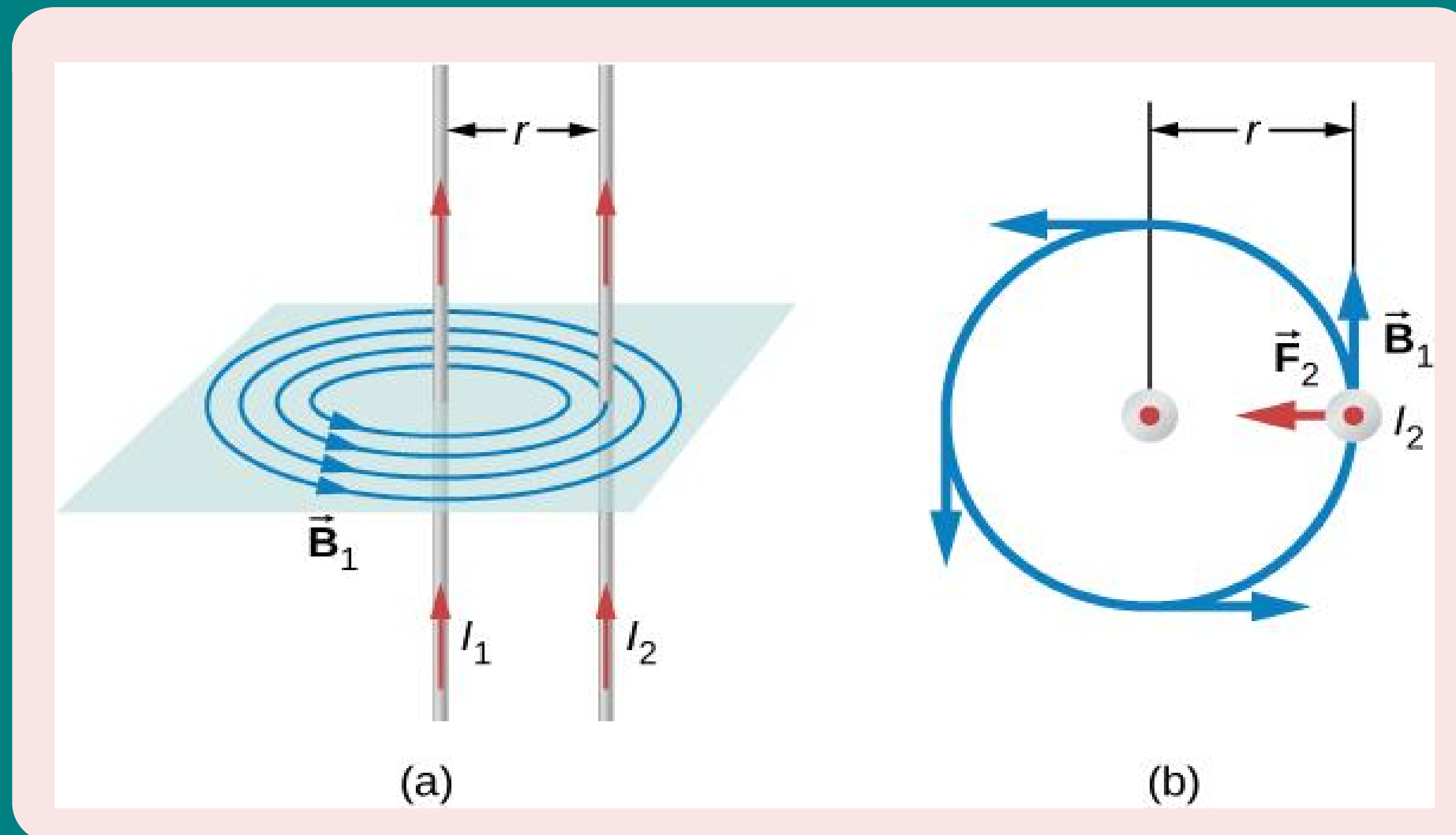


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- This **field (B_1)** produced by wire 1 **exert a force (F_{21}) on a length l of wire 2** and is given by.

FORCE BETWEEN TWO PARALLEL CURRENTS



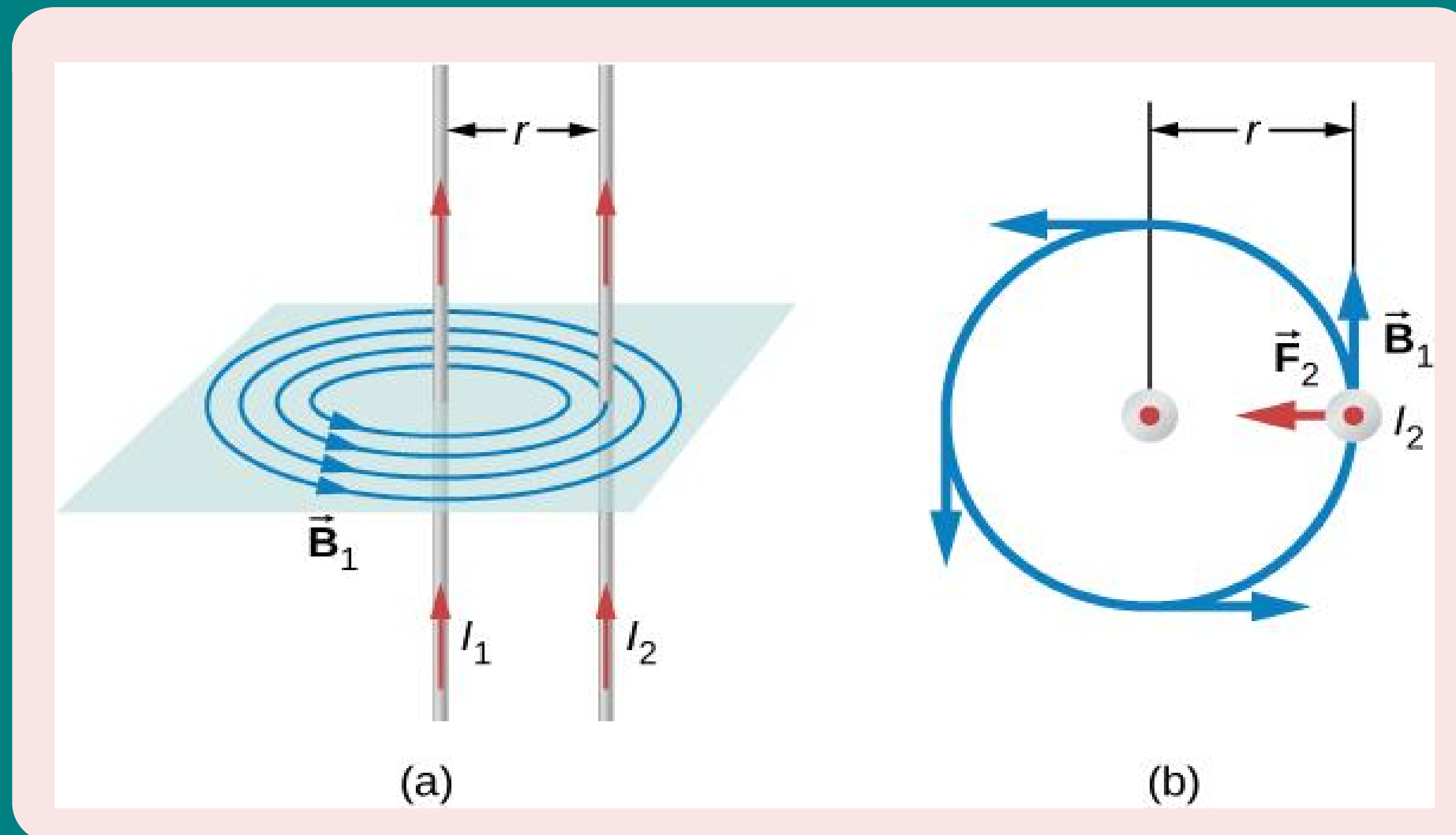
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$$F_{21} = I_2 l B_1$$

FORCE BETWEEN TWO PARALLEL CURRENTS



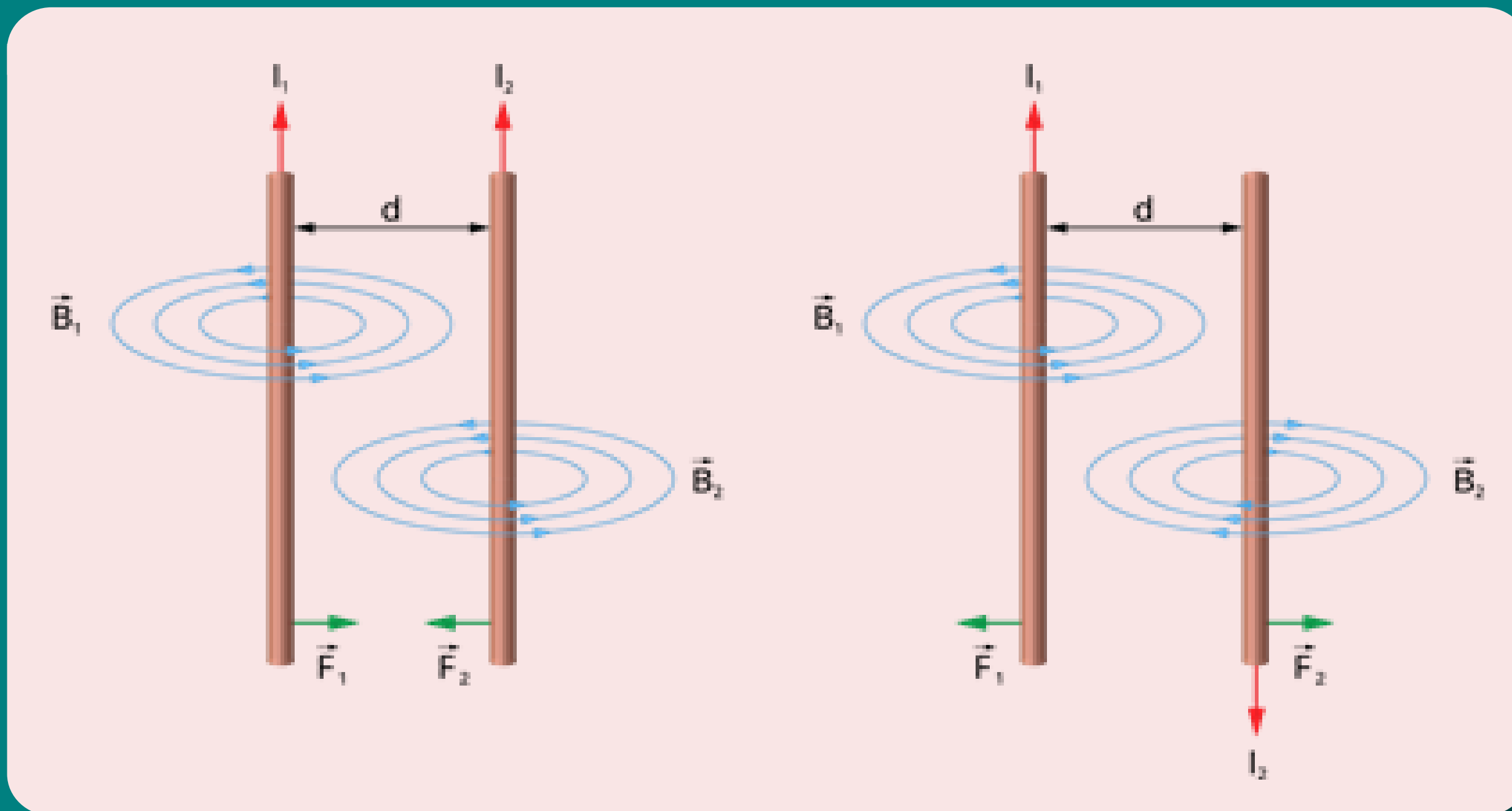
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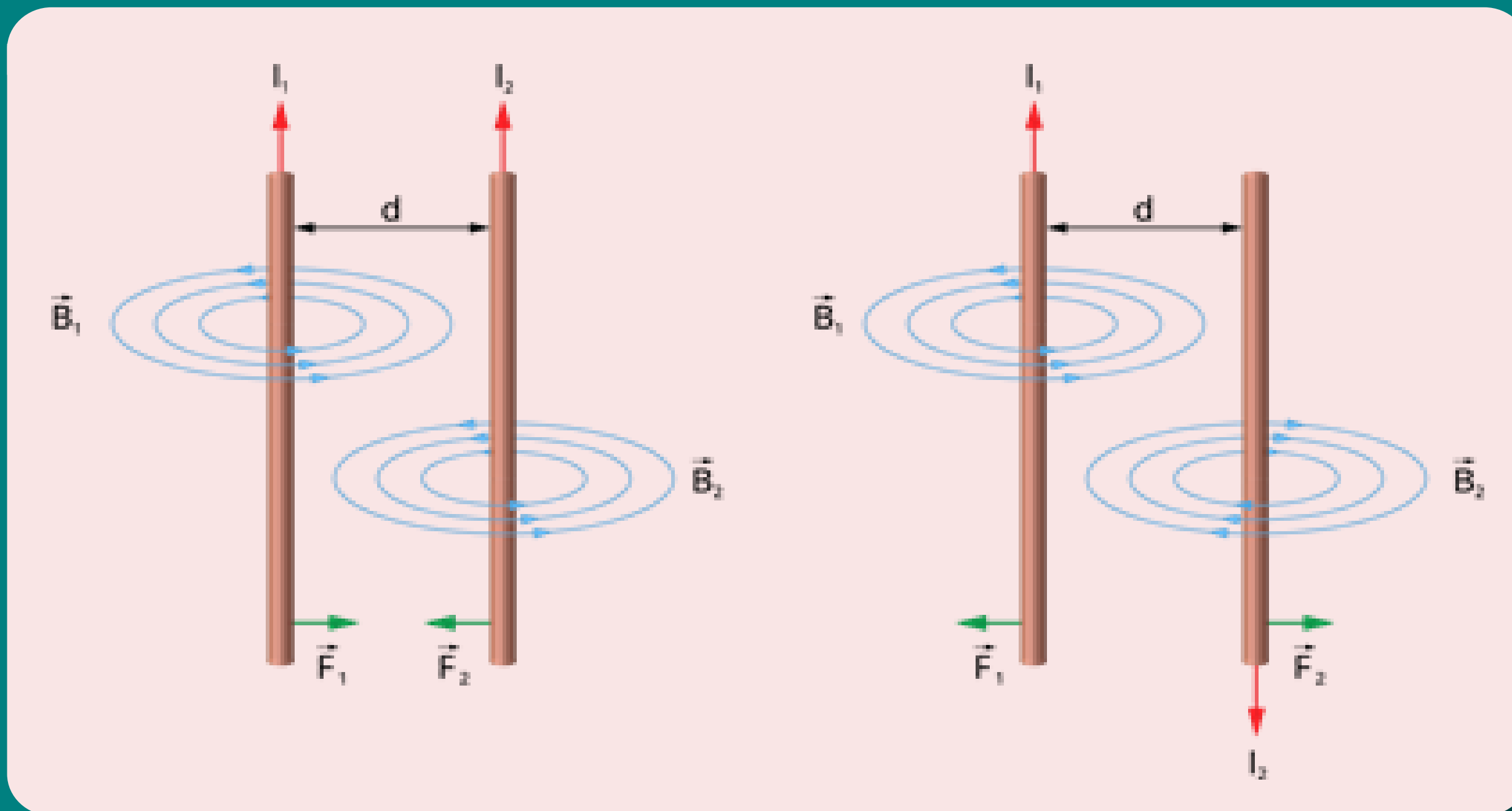
$$F_{21} = I_2 l B_1$$
$$F_{21} = \frac{\mu_0 I_1 I_2 l}{2\pi r} \dots\dots (2)$$

FORCE BETWEEN TWO PARALLEL CURRENTS



- Similarly, **field** (B_2) produced by wire 2 **exert a force** (F_{12}) **on a length** l **of wire 1** and its direction is opposite to F_{21}

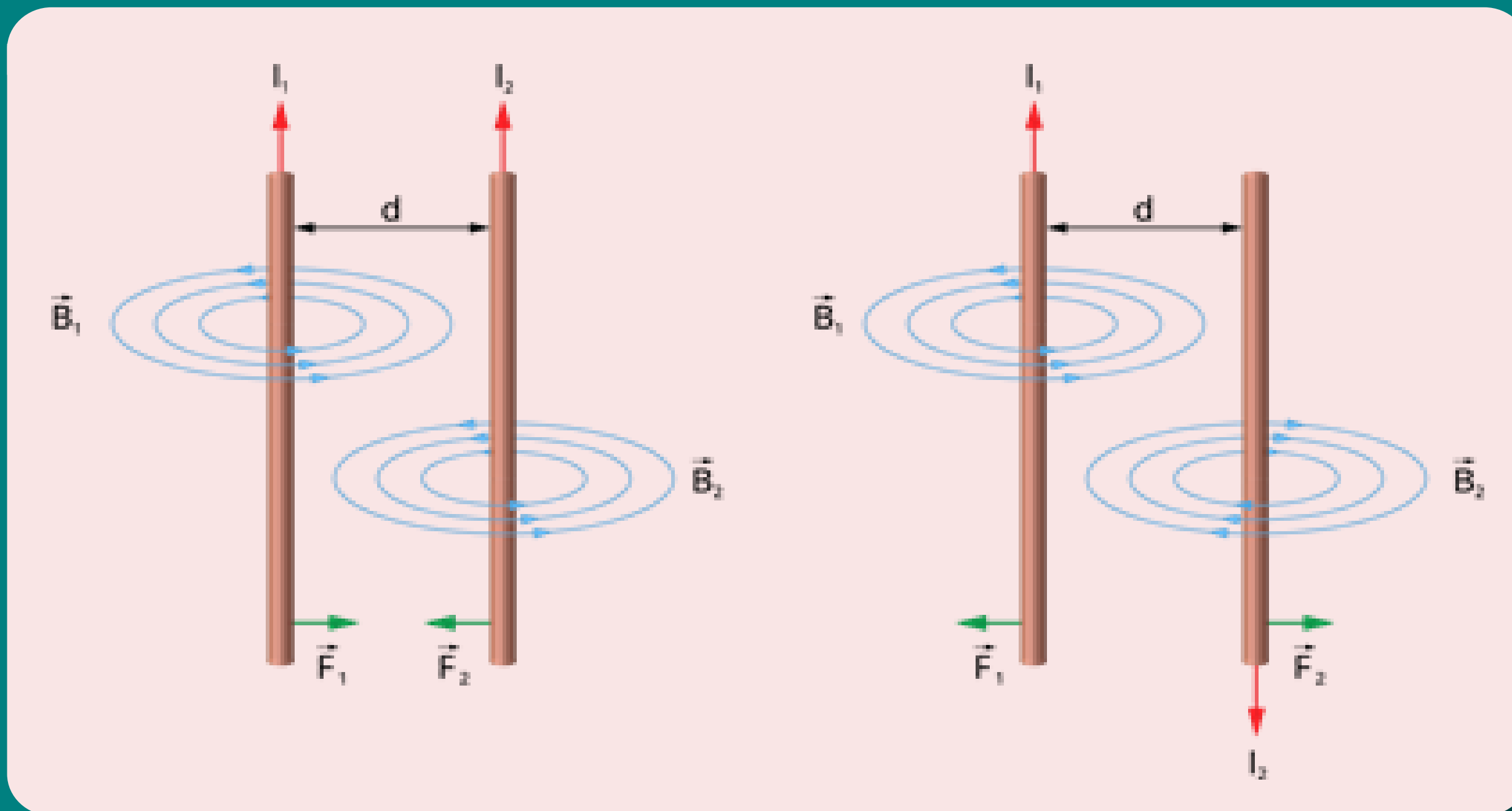
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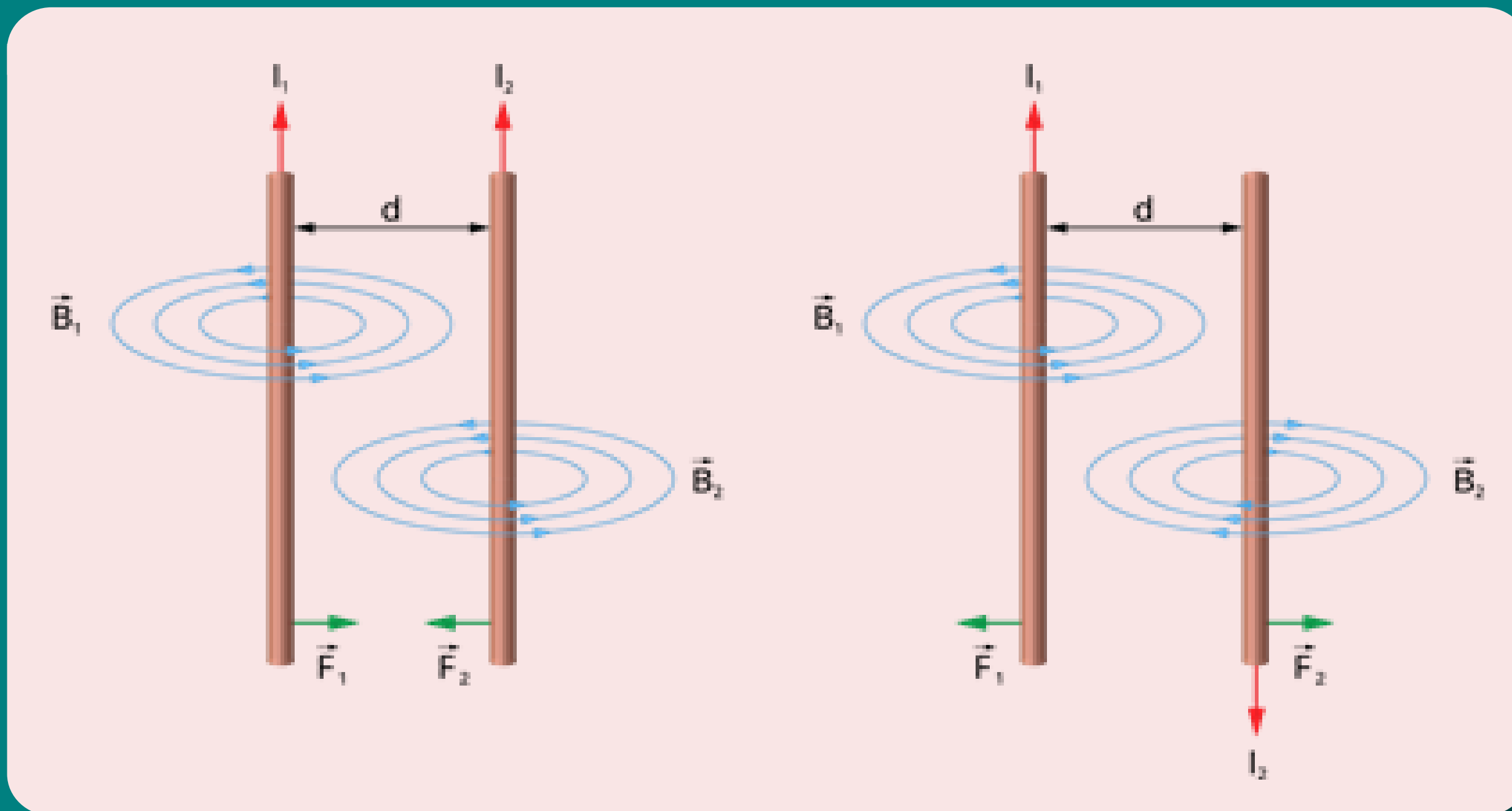
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FORCE BETWEEN TWO PARALLEL CURRENTS

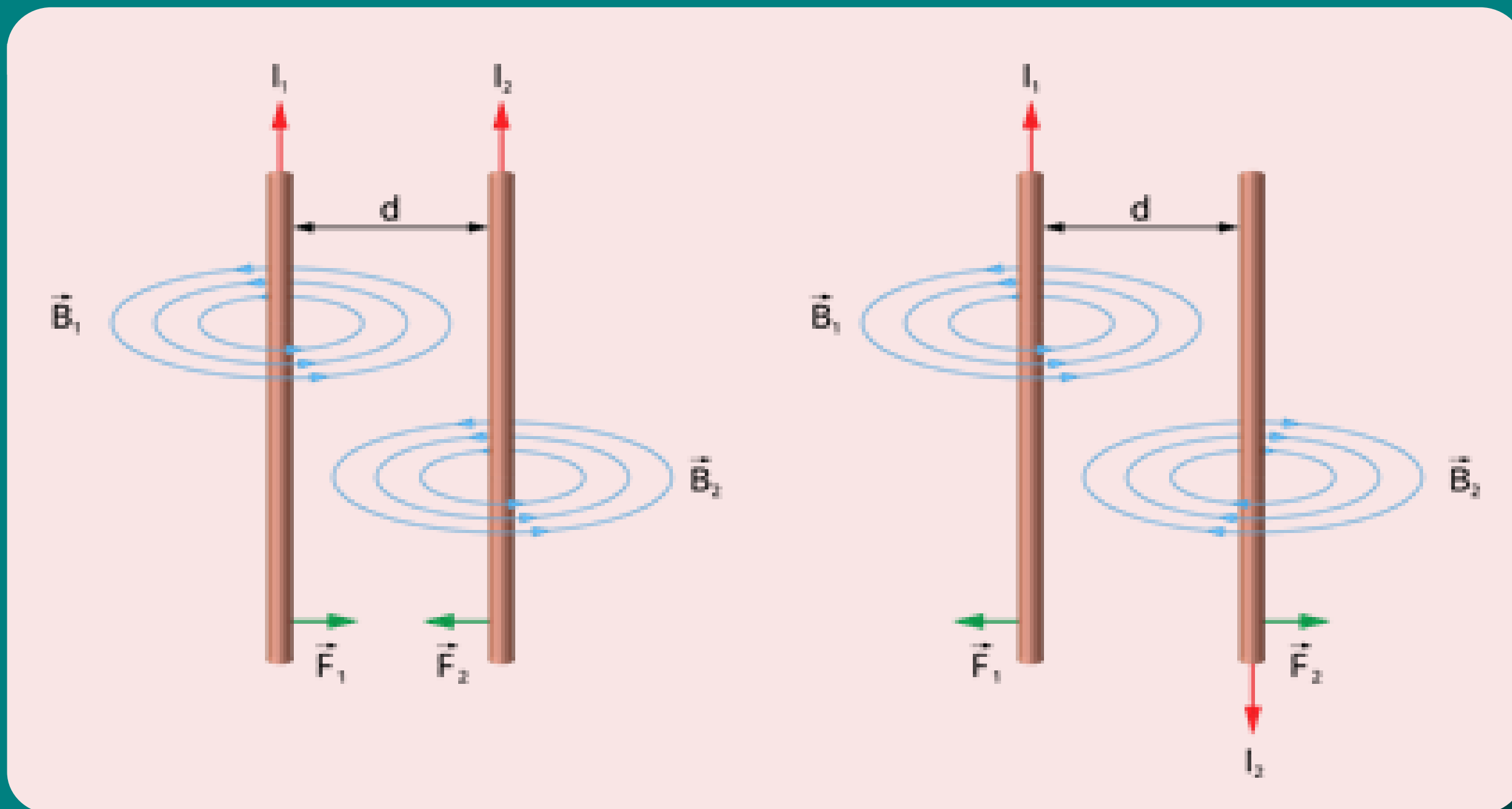


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- **Parallel currents (same direction) attract, and antiparallel currents (opposite direction) repel.**

FORCE BETWEEN TWO PARALLEL CURRENTS

Definition of S.I. Unit of current, Ampere

- The **definition of the ampere is based on the force between current - carrying wires**. Note that for long, parallel wires separated by **1 meter** with each carrying **1 ampere**, the **force per meter** is

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$$\frac{F}{l} = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{r}$$

$$\frac{F}{l} = \left(10^{-7} \frac{T \cdot m}{A} \right) \frac{2 A^2}{1 m} = 2 \times 10^{-7} N/m$$

Exercise 4.7

Two long and parallel straight wires A and B carrying currents of 8.0 A and 5.0 A in the same direction are separated by a distance of 4.0 cm. Estimate the force on a 10 cm section of wire A.

- 32.** (a) (i) Two long parallel straight conductors carrying current I_1 and I_2 are kept r distance apart in air. Obtain an expression for the force per unit length on one conductor due to the magnetic field produced by the other conductor. Hence, define one ampere. Under what condition will the force between the conductors be attractive in nature ?
- (ii) A closely wound circular coil of radius 6.28 cm, having 50 turns carries a steady current of 4 A. Find the magnetic field at the centre of the coil. How will the magnitude of the magnetic field be affected if the radius of the coil is halved keeping other factors same ?

(i) Obtaining expression of force/length	$1\frac{1}{2}$
Definition of one ampere	1
Condition of force to be attractive	$\frac{1}{2}$
(ii) Calculation of magnetic field	$1\frac{1}{2}$
Effect on magnitude of magnetic field when radius is halved	$\frac{1}{2}$

(i) Consider two long parallel conductors P and Q, distance r apart carrying currents I_1 and I_2 respectively. magnetic field of P at a point in the position of Q $B_1 = \frac{\mu_o I_1}{2\pi r} \text{ (i)}$ Magnetic force on Q at length l $F_{QP} = B_1 I_2 l = \frac{\mu_o I_1 I_2}{2\pi r} l \text{ (ii)}$ $f_2 = \frac{F_{QP}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \text{ (iii)}$ Similarly force per unit length of P due to magnetic field of Q $f_1 = \frac{F_{PQ}}{l} = \frac{\mu_o I_1 I_2}{2\pi r} \text{ (iv)}$ From eq. (ii) and (iv) $f_1 = f_2 = f = \frac{\mu_o I_1 I_2}{2\pi r} \text{ (v)}$ Definition of one ampere One ampere is the value of the steady current which when maintained in each of the two very long straight parallel conductors of negligible cross section, placed one meter apart in vaccum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of length. The force between the two conductors will be attractive when they carry current in the same direction.		(ii) Magnetic field at the centre of the current carrying coil $B = N \frac{\mu_o I}{2a}$ a= 6.28 cm N = 50 I = 4A $B = \frac{50 \times 4\pi \times 10^{-7} \times 4}{2 \times 6.28 \times 10^{-2}}$ $= 4 \times 10^{-3} \text{ T}$ When radius of the coil is halved i.e. $\frac{a}{2}$ then magnetic field becomes two times. $B = 8 \times 10^{-3} \text{ T}$
	1/2	1/2
	1/2	1/2
	1/2	
	1	
	1/2	

29. Two infinitely long parallel conductors A and B, separated by 2 cm, carry 6 A and 2 A currents in opposite directions respectively. Find :

- (a) the net magnetic field at a point midway between A and B.
- (b) the force acting per unit length on B.

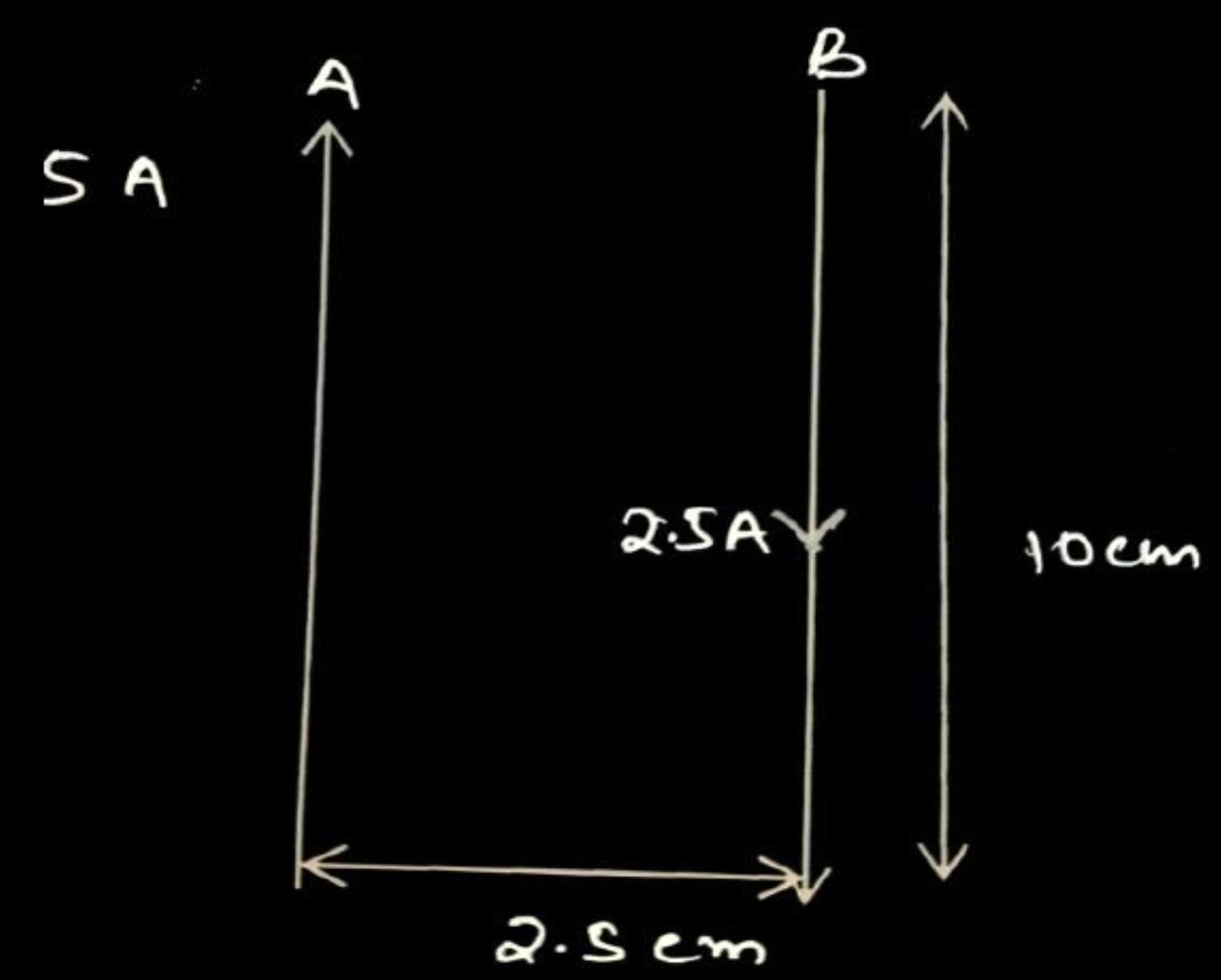
3

(a) Calculation of magnetic field due to current in A and B	$\frac{1}{2} + \frac{1}{2}$
Net magnetic field	1
(b) Force per unit length	1

(a) $B_1 = \frac{\mu_0 I_1}{2\pi r} = \frac{2 \times 10^{-7} \times 6}{1 \times 10^{-2}}$ $= 1.2 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$B_2 = \frac{\mu_0 I_2}{2\pi r} = \frac{2 \times 10^{-7} \times 2}{1 \times 10^{-2}}$ $= 0.4 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2}$
$\therefore B_1$ and B_2 are in same direction.	
$B_{\text{net}} = B_1 + B_2$ $= 1.2 \times 10^{-4} \text{ T} + 0.4 \times 10^{-4} \text{ T}$	
$= 1.6 \times 10^{-4} \text{ T}$ into the page	$\frac{1}{2} + \frac{1}{2}$
(b) Magnitude of magnetic force per unit length	
$\frac{F}{l} = \frac{\mu_0}{2\pi} \frac{I_1 I_2}{r}$	$\frac{1}{2}$
$= \frac{2 \times 10^{-7} \times 6 \times 2}{2 \times 10^{-2}} = 1.2 \times 10^{-4} \text{ N/m}$	$\frac{1}{2}$

- (b) (i) Derive an expression for the force acting on a current carrying straight conductor kept in a magnetic field. State the rule which is used to find the direction of this force. Give the condition under which this force is (1) maximum, and (2) minimum.
- (ii) Two long parallel straight wires A and B are 2.5 cm apart in air. They carry 5.0 A and 2.5 A currents respectively in opposite directions. Calculate the magnitude of the force exerted by wire A on a 10 cm length of wire B.

i) Derivation of expression for force	2
Statement of Rule	$\frac{1}{2}$
Conditions for maximum and minimum force	$\frac{1}{2} + \frac{1}{2}$
ii) Calculation of magnitude of force	$1 \frac{1}{2}$

Consider a rod of uniform cross sectional area A and length l. Let the number density of mobile charge carriers in it be n. Thus the total number of mobile charge carriers in it is n l A. For steady current I, drift velocity of electrons $\vec{v_d}$, in the presence of external magnetic field $\vec{B}$, the force on these carriers is	1/2	ii)	
$\vec{F} = n l A q (\vec{v_d} \times \vec{B})$	1/2		
$= [\vec{j} A l] \times \vec{B}$	1/2		
$= I (\vec{l} \times \vec{B})$	1/2		
Where $nq\vec{v_d}$ is current density ($\vec{j}$) and $ \vec{j}A $ is current (I)			
Fleming's left hand Rule: If forefinger, middle finger and thumb are stretched in mutually perpendicular directions, such that forefinger indicates the direction of magnetic field, middle finger indicates the direction of current in the conductor, then thumb indicates the direction of force on the conductor.			
Or			
Right Hand Thumb Rule : If the fingers of right hand are made to rotate from $\vec{l}$ to $\vec{B}$ through angle θ , the thumb points in the direction of force on the current carrying conductor.	1/2		
Condition for maximum force $\theta = 90^\circ$	1/2		
$ \vec{F} = I l B \sin \theta = I l B$			1/2
Condition for minimum force $\theta = 0^\circ$ or 180°			
$ \vec{F} = 0$	1/2		1/2
			
		$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d} l$ $= \frac{10^{-7} \times 2 \times 5 \times 2.5}{2.5 \times 10^{-2}} \times 10 \times 10^{-2} \text{ N}$ $= 10^{-5} \text{ N}$	

13. Beams of electrons and protons move parallel to each other in the same direction. They

1

- (a) attract each other.
- (b) repel each other.
- (c) neither attract nor repel.
- (d) force of attraction or repulsion depends upon speed of beams.

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(b) Repel each other

2. Two long parallel wires kept 2 m apart carry 3A current each, in the same direction. The force per unit length on one wire due to the other is

1

(A) $4.5 \times 10^{-5} \text{ Nm}^{-1}$, attractive

(B) $4.5 \times 10^{-7} \text{ N/m}$, repulsive

(C) $9 \times 10^{-7} \text{ N/m}$, repulsive

(D) $9 \times 10^{-5} \text{ N/m}$, attractive

5. Two long straight parallel conductors carrying steady currents I_1 and I_2 in the same direction are separated by a distance d . The force exerted by one conductor on unit length of the other is :

(a) $\frac{\mu_0 I_1 I_2}{4\pi d}$, attractive

(b) $\frac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\frac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\frac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

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(b) $\frac{\mu_0 I_1 I_2}{4\pi d}$, repulsive

(c) $\frac{\mu_0 I_1 I_2}{2\pi d}$, attractive

(d) $\frac{\mu_0 I_1 I_2}{2\pi d}$, repulsive

(c) $\frac{\mu_o I_1 I_2}{2\pi d}$, attractive

- (b) (i) Derive an expression for the force acting on a conductor carrying current I in a magnetic field $\vec{B}$.
- (ii) A long horizontal conductor carries a current of 30 A, in east to west direction. What are the magnitude and direction of the magnetic field due to the current 1.0 m below the conductor ?

5

(i) Deriving expression for force on a current carrying wire

3

(ii) Finding the magnitude and direction of magnetic field

1+1

(i) Consider a wire of length l , area of cross- section A and having number density of free electrons n and carrying electric current I . Total number of charge carriers in the wire = nIA . If $\vec{v}_d$ is drift velocity of electrons, magnetic force experience by the electrons is $\vec{F} = -(nIA)e\vec{v}_d \times \vec{B}$ As $I\vec{l} = -enIA\vec{v}_d$ $\vec{F} = I(\vec{l} \times \vec{B})$	$\frac{1}{2}$
(ii) $B = \frac{2\mu_o I}{4\pi r}$	$\frac{1}{2}$
$B = \frac{2 \times 10^{-7} \times 30}{1} = 0.6 \times 10^{-5} T$	$\frac{1}{2}$
Magnetic field is directed towards SOUTH	1

17. Derive an expression for magnetic force $\vec{F}$ acting on a straight conductor of length L carrying current I in an external magnetic field $\vec{B}$. Is it valid when the conductor is in zig-zag form ? Justify.

Deriving an expression for magnetic force	$1\frac{1}{2}$
Validity and Justification for zig-zag form conductor	$\frac{1}{2}$

Total number of mobile charge carriers in a conductor of length L , cross-sectional area A and number density of charge carriers n :

$$= nLA$$

Force acting on the charge carriers in external magnetic field $\vec{B}$

$$\vec{F} = (nAL) q \vec{v}_d \times \vec{B} \quad \text{-----}(1)$$

1/2

Where $\vec{v}_d$ is the drift velocity of the charge carriers

Current flowing

$$I = v_d q n A$$

1/2

$$I \vec{L} = \vec{v}_d q n A L \quad \text{-----}(2)$$

On solving equation (1) and (2)

$$\vec{F} = I(\vec{L} \times \vec{B})$$

1/2

Yes, because this force can be calculated by considering zig-zag conductor as a collection of linear strips ($d\vec{l}$) and summing them vectorically.

1/2

5. Two long straight parallel conductors A and B, kept at a distance r , carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is

1

(A) $\frac{3\mu_0 I I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I I_1}{4\pi r}$, towards B

5. Two long straight parallel conductors A and B, kept at a distance r , carry current I in opposite directions. A third identical conductor C, kept at a distance $\left(\frac{r}{3}\right)$ from A carry current I_1 in the same direction as in A. The net magnetic force on unit length of C is

1

(A) $\frac{3\mu_0 I I_1}{2\pi r}$, towards A

(B) $\frac{3\mu_0 I I_1}{2\pi r}$, towards B

(C) $\frac{3\mu_0 I I_1}{4\pi r}$, towards A

(D) $\frac{3\mu_0 I I_1}{4\pi r}$, towards B

No option is correct, award 1 mark.

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

- (A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.
- (D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

4. Two thin long parallel wires separated by a distance 'a' carry current 'I' in opposite directions. The wires will :

- (A) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (B) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a^2}$, per unit length.
- (C) Attract each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.
- (D) Repel each other with a force $\frac{\mu_0 I^2}{2\pi a}$, per unit length.

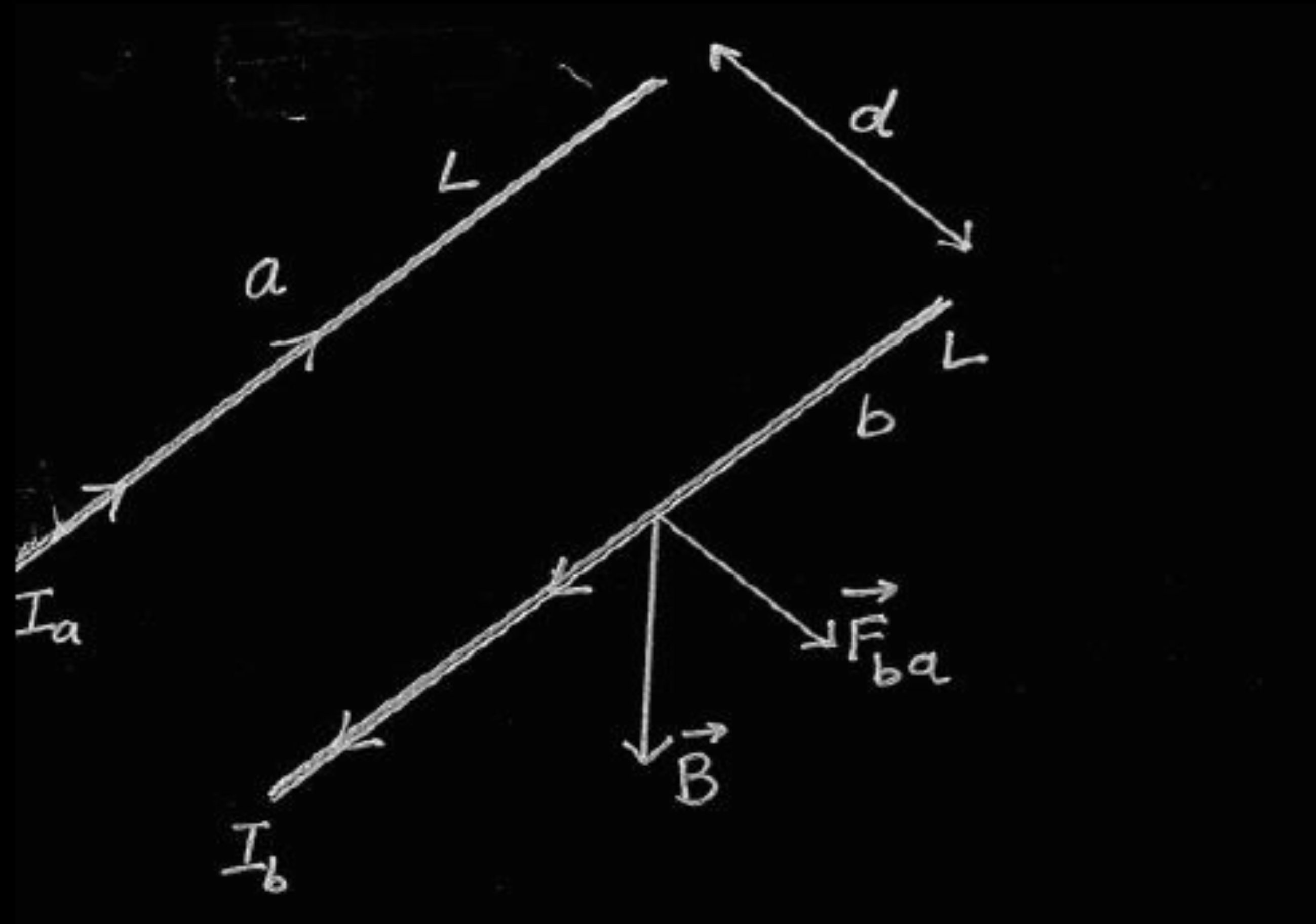
(D) Repel each other with a force $\frac{\mu_o I^2}{2\pi a}$, per unit length

23. Two long straight parallel conductors carrying currents, exert a force on each other. Why ? Derive an expression for the force per unit length between two long straight parallel conductors carrying currents in opposite directions. Explain the nature of the force between these conductors.

3

Reason for exerting force on straight parallel conductors	$\frac{1}{2}$
Derivation for force per unit length	2
Explanation of nature of Force	$\frac{1}{2}$

One conductor experiences a force due to magnetic field of the other conductor



1/2

1/2

Magnetic field produced by conductor 'a' at all points along the length of conductor 'b'

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

1/2

Force on conductor 'b' due to this magnetic field

$$F_{ba} = I_b L B_a$$

1/2

$$F_{ba} = \frac{\mu_0 I_a I_b L}{2\pi d}$$

$$f_{ba} = \frac{F_{ba}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$$

directed away from a

1/2

$$f_{ab} = \frac{F_{ab}}{L} = \frac{\mu_0 I_a I_b}{2\pi d}$$

directed away from b

Repulsive, the forces acting on them are away from each other.

1/2

- 22.** (a) Two long, straight, parallel conductors carry steady currents in opposite directions. Explain the nature of the force of interaction between them. Obtain an expression for the magnitude of the force between the two conductors. Hence define one ampere. 3

OR

- (b) Obtain an expression for the torque $\vec{\tau}$ acting on a current carrying loop in a uniform magnetic field $\vec{B}$. Draw the necessary diagram. 3

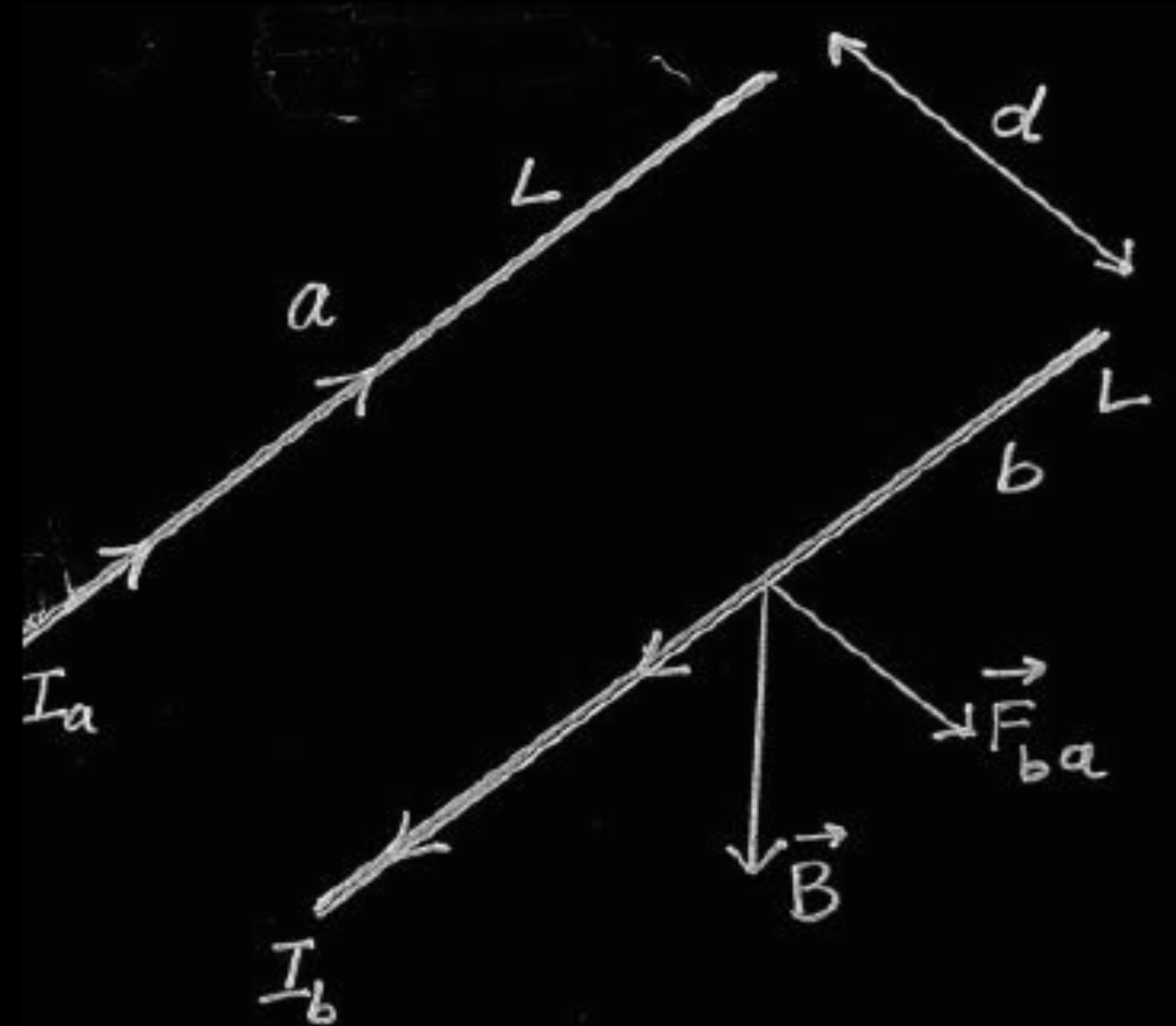
(a)

Explaining nature of force	$\frac{1}{2}$
Obtaining expression of force	$1\frac{1}{2}$
Defining one ampere	1

(b)

Obtaining expression of torque	2
Drawing diagram	1

Nature of force is repulsive.



Magnetic field due to current I_a at all points of conductor b

$$B_{ab} = \frac{\mu_0 I_a}{2\pi d} \quad \text{directed downwards}$$

Force experienced by conductor b on its segment of length l

$$F_{ab} = I_b l B_{ab}$$

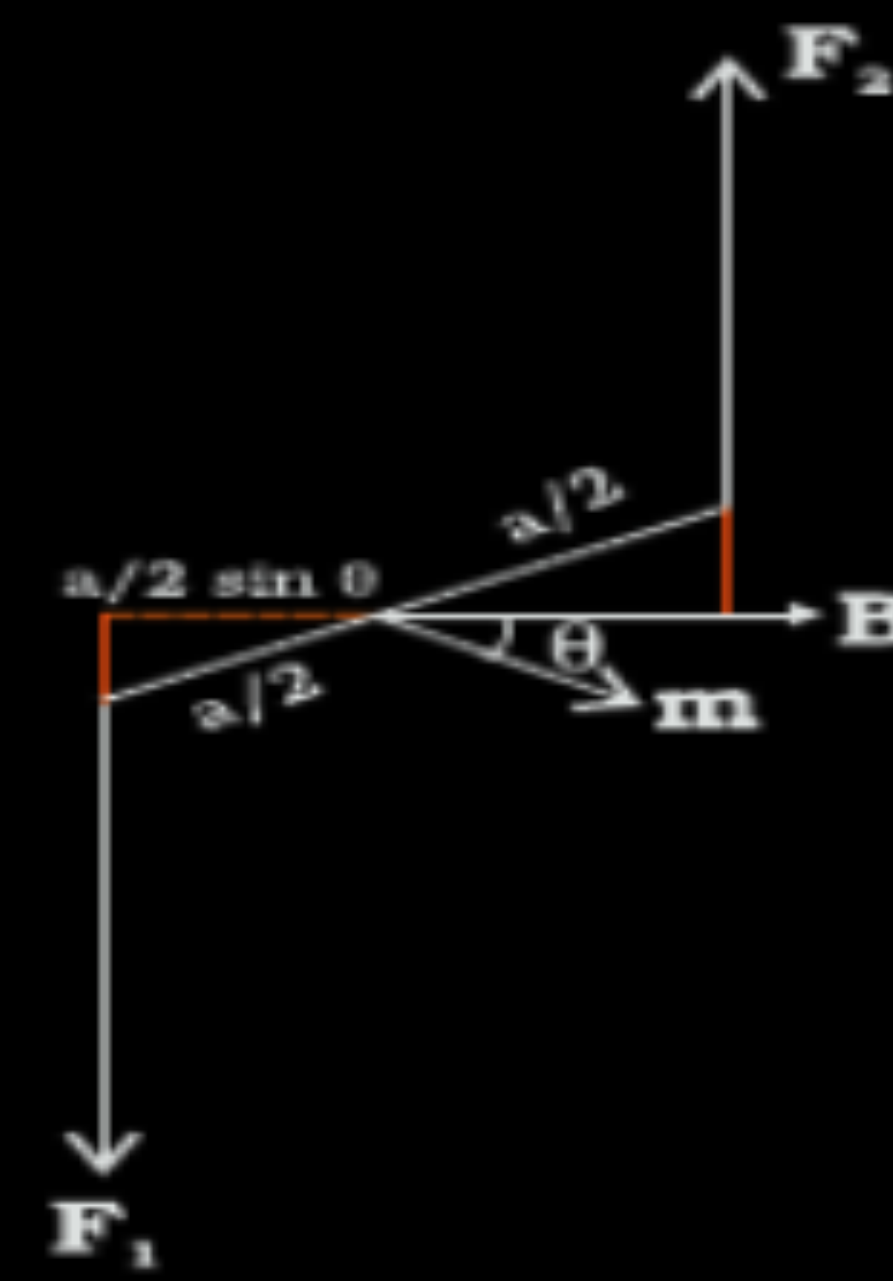
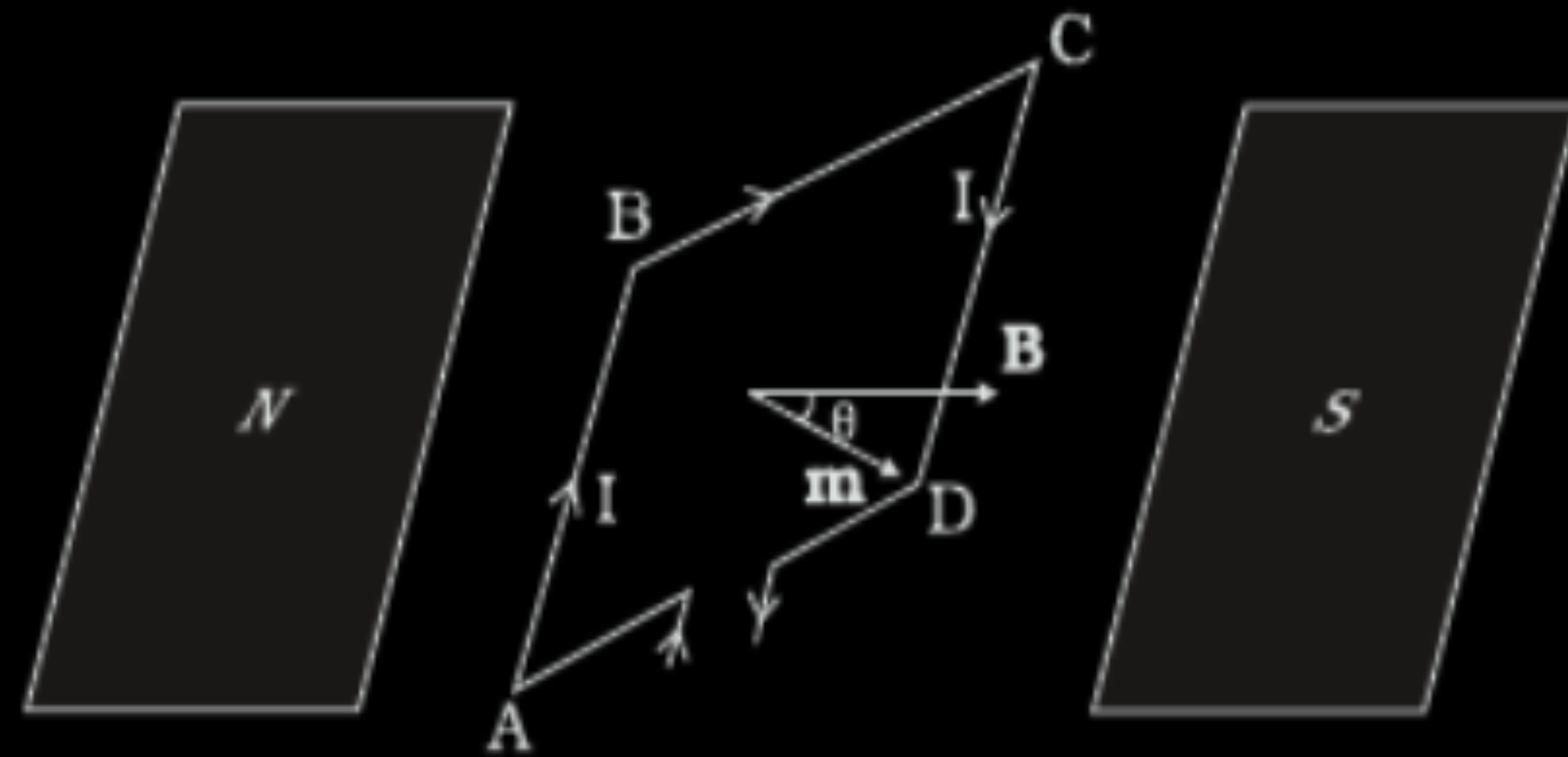
$$= \frac{\mu_0 I_a I_b}{2\pi d} l \quad \text{directed towards left}$$

Similarly

Force experienced by conductor a on its segment of length l

$$F_{ba} = \frac{\mu_0 I_a I_b}{2\pi d} l \quad \text{directed towards right}$$

One ampere is that steady current which when maintained in each of two very long straight parallel conductors of negligible cross-section, placed one metre apart in vacuum produces a force of 2×10^{-7} N/m on each conductor.



1

$\frac{1}{2}$

Forces on arm BC and DA are equal and opposite and act along the axis of the coil. Being collinear they cancel each other.

Forces on arms AB and CD are equal and opposite but not collinear. They form a couple.

$$F_1 = F_2 = I b B$$

$$\tau = F_1 \frac{a}{2} \sin \theta + F_2 \frac{a}{2} \sin \theta$$

$\frac{1}{2}$

$$\tau = I a b B \sin \theta$$

$$\tau = I A B \sin \theta \quad (\text{where } A = ab \text{ \& } m = IA)$$

$$\vec{\tau} = \vec{m} \times \vec{B}$$

$\frac{1}{2}$

6. A 10 cm long wire lies along y-axis. It carries a current of 1.0 A in positive y-direction. A magnetic field $\vec{B} = (5 \text{ mT})\hat{j} - (8 \text{ mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0.8 \text{ mN})\hat{i}$

(B) $-(0.8 \text{ mN})\hat{i}$

(C) $(80 \text{ mN})\hat{i}$

(D) $-(80 \text{ mN})\hat{i}$

6. A 10 cm long wire lies along y-axis. It carries a current of 1.0 A in positive y-direction. A magnetic field $\vec{B} = (5 \text{ mT})\hat{j} - (8 \text{ mT})\hat{k}$ exists in the region. The force on the wire is :

(A) $(0.8 \text{ mN})\hat{i}$

(B) $-(0.8 \text{ mN})\hat{i}$

(C) $(80 \text{ mN})\hat{i}$

(D) $-(80 \text{ mN})\hat{i}$

(B) $(-0.8 \text{ mN})\hat{i}$

13. *Assertion (A) :* Two long parallel wires, freely suspended and connected in series to a battery, move apart.

Reason (R) : Two wires carrying current in opposite directions repel each other.

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(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A)

15. *Assertion (A) :* When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

15. *Assertion (A) :* When current is passed through a spring, two adjacent turns move towards each other.

Reason (R) : There is a force between any two adjacent turns of a current carrying spring.

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

- 23.** Two parallel straight wires of lengths 15 m and 20 m carrying currents 6.0 A and 8.0 A respectively in opposite directions, are 10 cm apart. Calculate the force on a 10 cm long section of the second wire near its centre. Also mention the nature of the force between these wires.

3

Calculation of force on wire

2

Nature of the force

1

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

$$= \frac{4\pi \times 10^{-7} \times 6 \times 8}{2\pi \times 10 \times 10^{-2}}$$

$$= 96 \times 10^{-6} \text{ N/m}$$

Force on 10 cm long section of II wire

$$F = 96 \times 10^{-6} \times 10 \times 10^{-2}$$

$$= 96 \times 10^{-7} \text{ N}$$

Nature of force is repulsive.

1

1/2

1/2

1

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

(A) 30°

(B) 45°

(C) 60°

(D) 90°

7. A current of 5 A is flowing in a wire of length 1.5 cm. A force of 7.5 mN acts on it when it is placed in a uniform magnetic field of 0.2 T. The angle between the magnetic field and the direction of current is :

(A) 30°

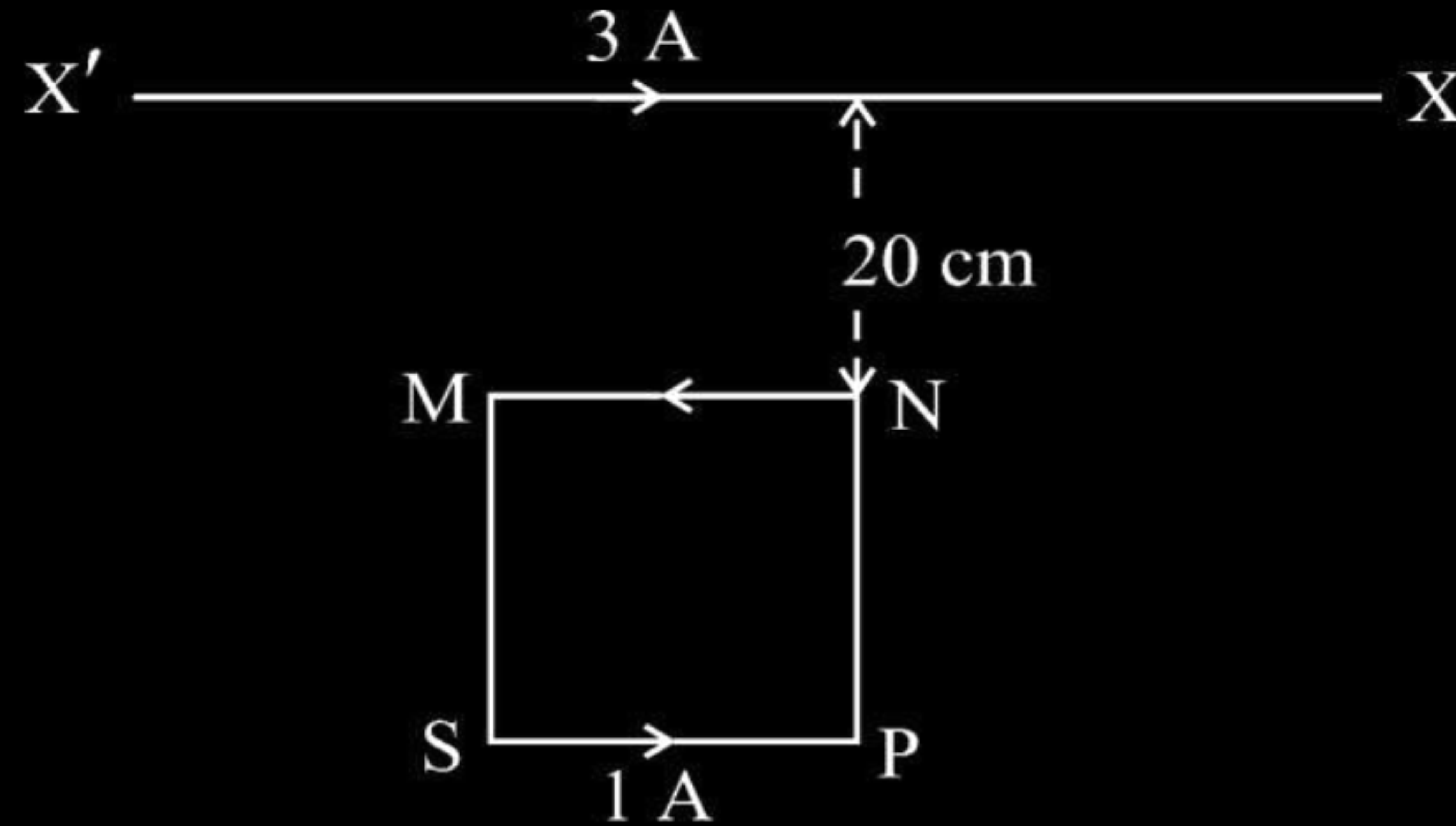
(B) 45°

(C) 60°

(D) 90°

(A) 30°

- (ii) A long, straight horizontal wire $X'X$ is held stationary and carries a current of 3.0 A . A square loop $MNPS$ of side 10 cm , carrying a current of 1.0 A is kept near the wire $X'X$ as shown in the figure. Find the magnitude and direction of the net magnetic force acting on the loop due to the wire.



7. A straight wire of length 1.0 m is placed along x-axis, in a region with magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. A current of 2.0 A flows in the wire along +x direction. The magnetic force acting on the wire is :

(A) 2.0 N, along z-axis

(B) 2.0 N, along -z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along -z-axis

7. A straight wire of length 1.0 m is placed along x-axis, in a region with magnetic field $\vec{B} = (3\hat{i} + 2\hat{j})$ T. A current of 2.0 A flows in the wire along +x direction. The magnetic force acting on the wire is :

(A) 2.0 N, along z-axis

(B) 2.0 N, along -z-axis

(C) 4.0 N, along z-axis

(D) 4.0 N, along -z-axis

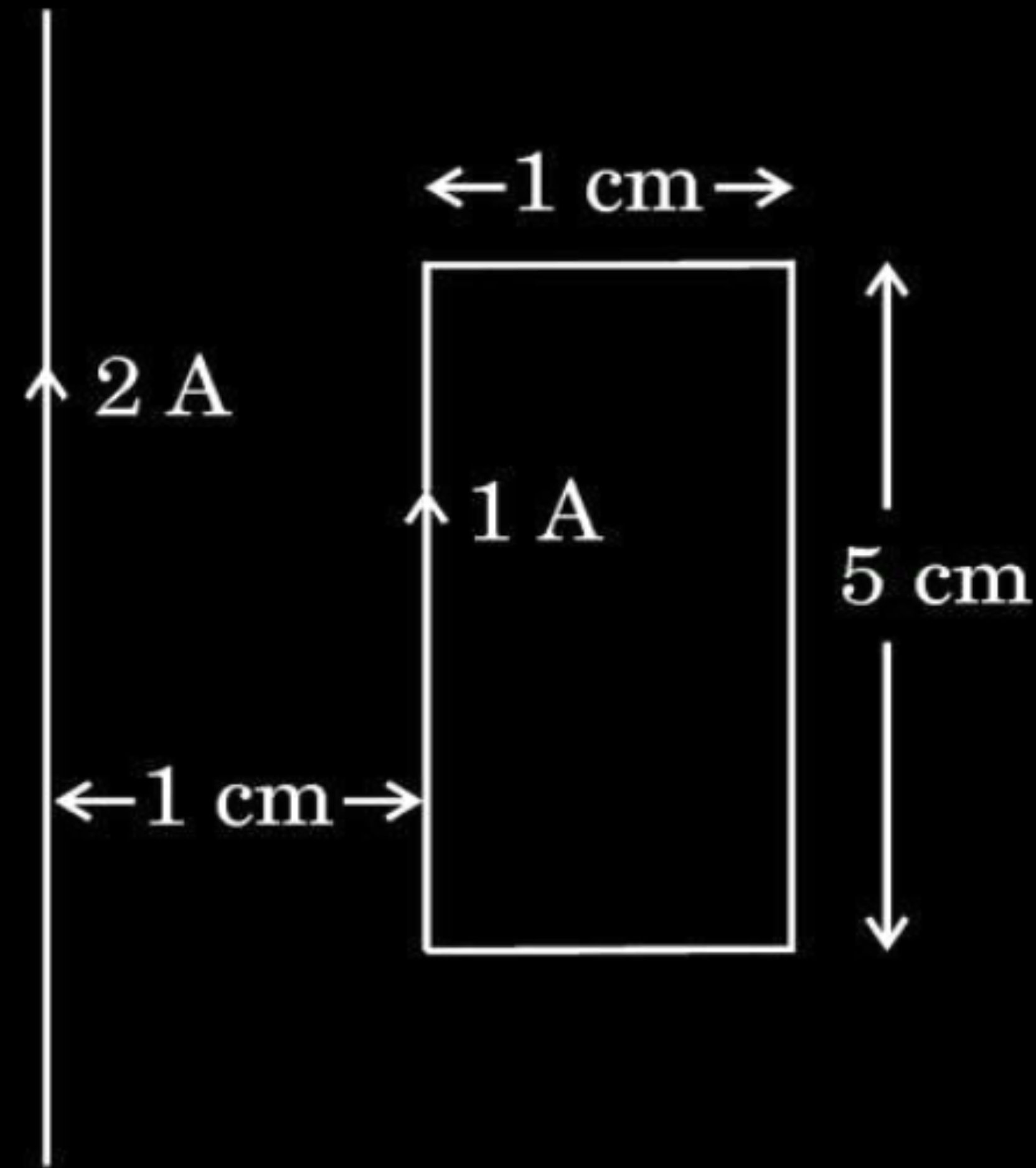
(C) 4.0 N, along z-axis

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x . The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :
- (A) become four times
 - (B) become two times
 - (C) becomes one-fourth
 - (D) remain the same

5. Two parallel straight wires carrying currents I_1 and I_2 are separated by a distance x . The current in the first wire is increased from I_1 to $2I_1$ and their separation is reduced to $\frac{x}{2}$. The value of force between the wires will :
- (A) become four times
 - (B) become two times
 - (C) becomes one-fourth
 - (D) remain the same

(A) become four times

24. A rectangular loop carries a current of 1 A. A straight long wire carrying 2 A current is kept near the loop in the same plane as shown in the figure.



Find :

- (i) the torque acting on the loop, and
- (ii) the magnitude and direction of the net force on the loop.

Finding-

- (i) The torque acting on the loop
- (ii) The magnitude and direction of net force

1

2

(i) $\tau = mB \sin\theta$	1/2
As $\vec{m}$ and $\vec{B}$ are in same direction, $\theta=0^\circ$	
$\tau=0$	1/2
(ii) $F = \frac{\mu_0 I_1 I_2 l}{2\pi r}$	1/2
$F_{net} = \frac{\mu_0 I_1 I_2 l}{2\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$ $= \frac{4\pi \times 10^{-7} \times 2 \times 1 \times 5 \times 10^{-2}}{2\pi \times 10^{-2}} \left(1 - \frac{1}{2} \right)$	1/2
$F_{net} = 1 \times 10^{-6} \text{ N}$	1/2
Net force on the loop is towards the long straight wire.	1/2

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0.6\hat{j} + 0.8\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

(A) $(0.12\hat{j} - 0.16\hat{k})$ N

(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

(C) $(-0.12\hat{j} + 0.16\hat{k})$ N

(D) $(0.16\hat{j} - 0.12\hat{k})$ N

3. A straight conductor is carrying a current of 2 A in +x direction along it. A uniform magnetic field $\vec{B} = (0.6\hat{j} + 0.8\hat{k})$ T is switched on, in the region. The force acting on 10 cm length of the conductor is :

(A) $(0.12\hat{j} - 0.16\hat{k})$ N

(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

(C) $(-0.12\hat{j} + 0.16\hat{k})$ N

(D) $(0.16\hat{j} - 0.12\hat{k})$ N

(B) $(-0.16\hat{j} + 0.12\hat{k})$ N

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is –

1

(A) zero

(B) 0.45 N

(C) 0.03 N

(D) 0.015 N

4. A 5.0 cm long wire carrying 1.0 A current is kept in a region in which a uniform magnetic field of 0.3 T is applied perpendicular to the length of the wire. The magnitude of the force acting on the wire is –

1

(A) zero

(B) 0.45 N

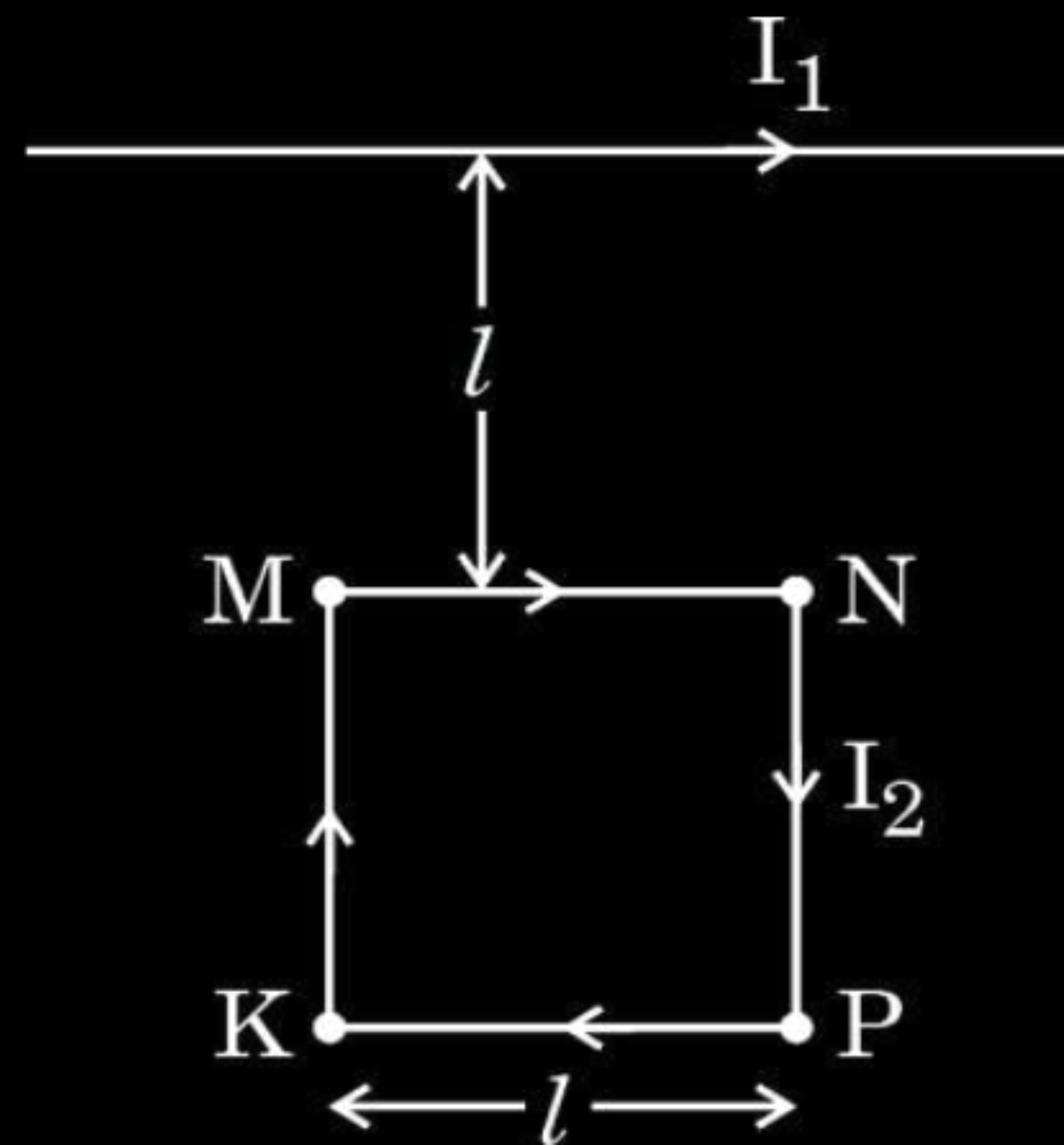
(C) 0.03 N

(D) 0.015 N

(D) 0.015 N

- 23.** (a) In which cases does a charged particle not experience a force in a magnetic field ?
- (b) A square loop MNPK of side ' l ' carrying a current ' I_2 ' is kept close to a long straight wire in the same plane and the wire carries a steady current I_1 as shown in the figure. Obtain the magnitude of magnetic force exerted by the wire on the loop.

3



- | | |
|--|---|
| (a) Conditions for, no force experienced by charged particle in magnetic field | 1 |
| (b) Obtaining the magnitude of magnetic force exerted by wire on the loop | 2 |

3. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4 \text{ mT}) \hat{j} + (0.6 \text{ mT}) \hat{k}$ is switched on, in the region. The force acting on the segment is

(A) $(2\hat{j} + 3\hat{k}) \text{ mN}$

(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

(C) $(6\hat{j} + 4\hat{k}) \text{ mN}$

(D) $(-4\hat{j} + 6\hat{k}) \mu\text{N}$

3. A 1 cm segment of a wire lying along x-axis carries current of 0.5 A along +x direction. A magnetic field $\vec{B} = (0.4 \text{ mT}) \hat{j} + (0.6 \text{ mT}) \hat{k}$ is switched on, in the region. The force acting on the segment is

1

(A) $(2\hat{j} + 3\hat{k}) \text{ mN}$

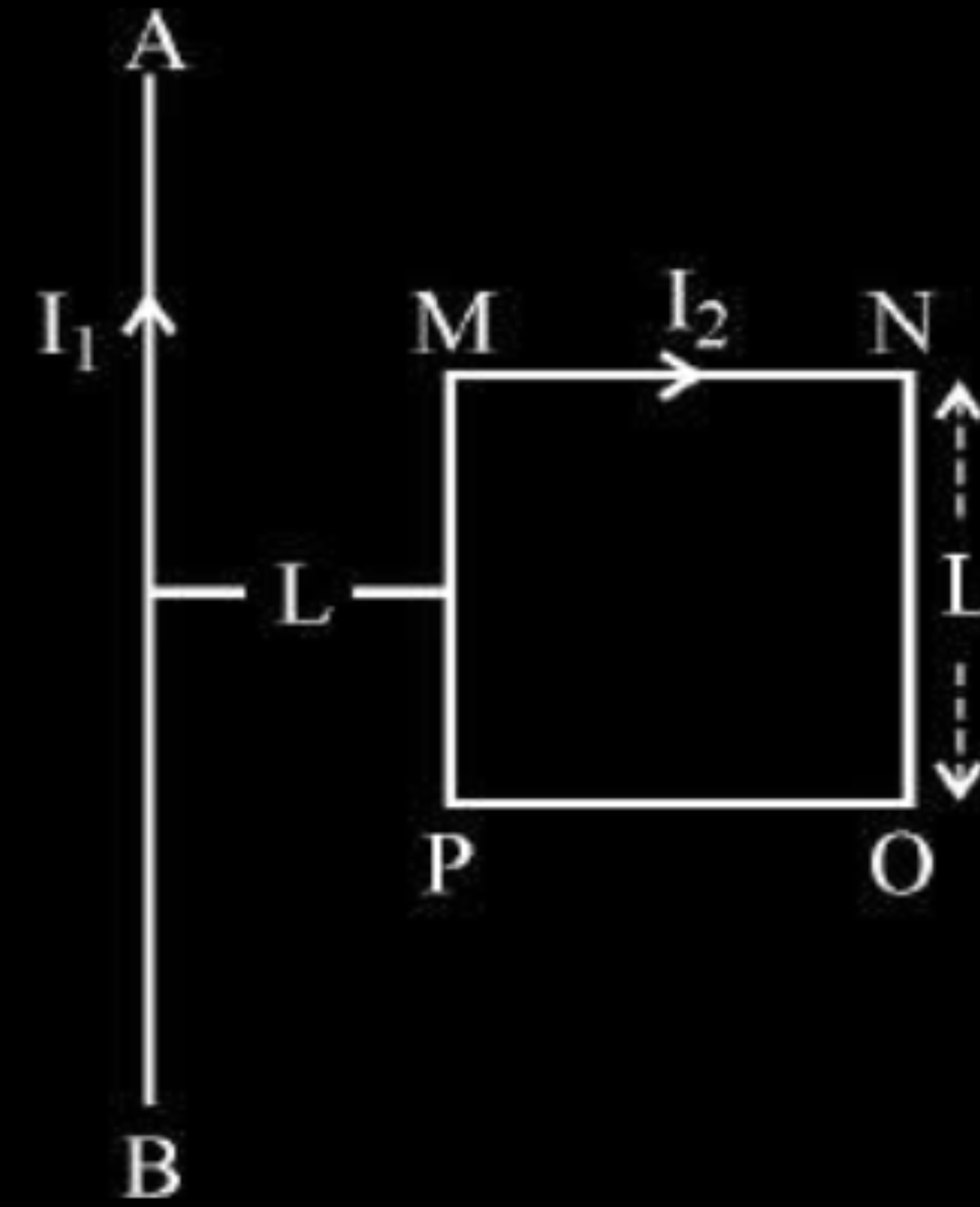
(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

(C) $(6\hat{j} + 4\hat{k}) \text{ mN}$

(D) $(-4\hat{j} + 6\hat{k}) \mu\text{N}$

(B) $(-3\hat{j} + 2\hat{k}) \mu\text{N}$

A square shaped current carrying loop MNOP is placed near a straight long current carrying wire AB as shown in the fig. The wire and the loop lie in the same plane. If the loop experiences a net force F towards the wire, find the magnitude of the force on the side 'NO' of the loop.



Magnitude of force on side NO is $= F$

Alternatively

Let force on side MP be $= F_1$

Force on side $NO = \frac{F_1}{2}$

Magnitude of net force $= F_1 - \frac{F_1}{2} = \frac{F_1}{2} = F$

Therefore force on side NO $= \frac{F_1}{2} = F$

Give full credit if a student calculates the force as shown below.

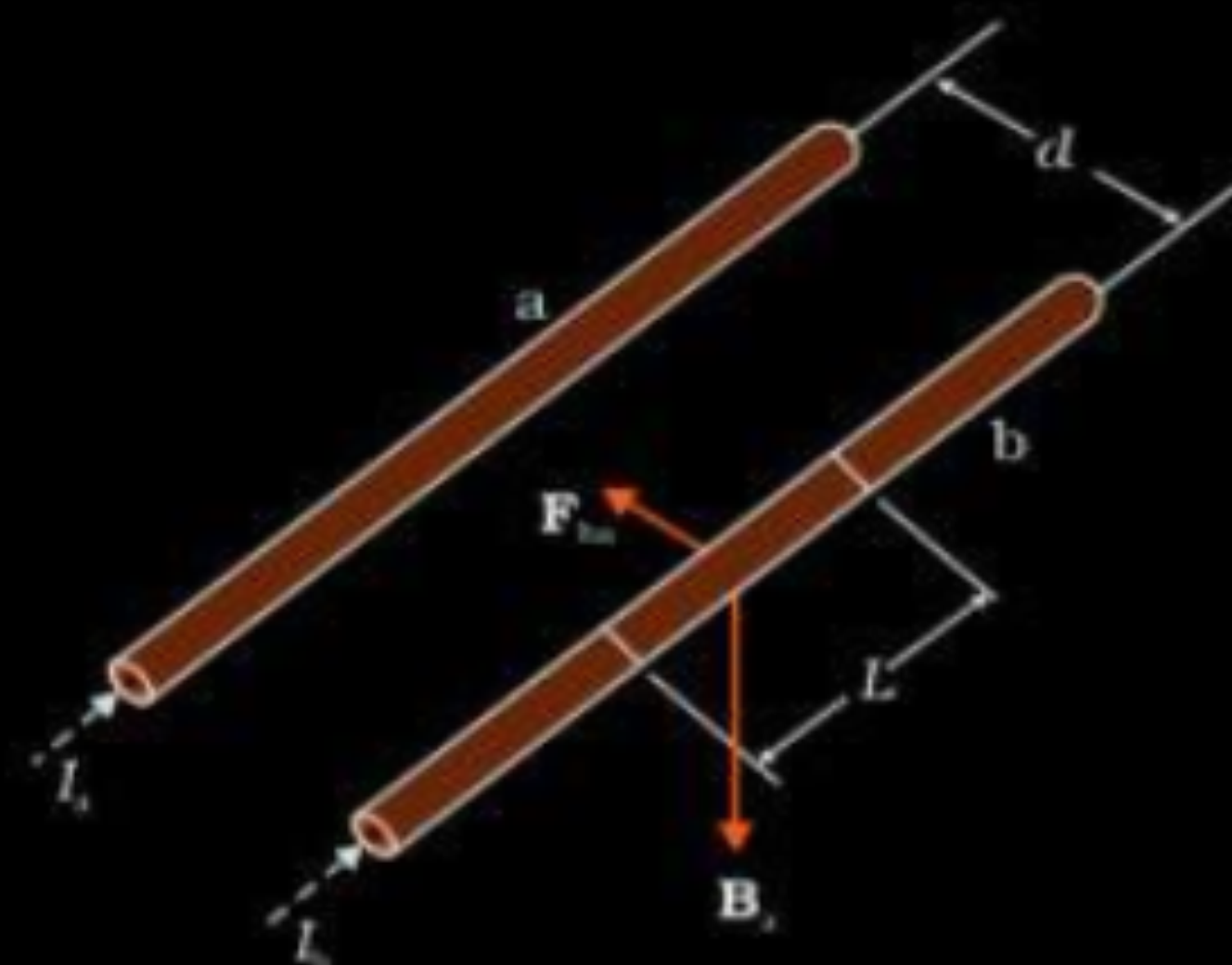
$$F = \frac{\mu_0}{2\pi} I_1 I_2$$

$\frac{1}{2}$

$\frac{1}{2}$

- (a) Derive the expression for the force acting between two long parallel current carrying conductors. Hence, define 1 A current.
- (b) A steady current of 2A flows through a circular coil having 5 turns of radius 7 cm. The coil lies in X-Y plane with its centre at the origin. Find the magnitude and direction of the magnetic dipole moment of the coil.

(a)



Magnetic field due to the current I_a flowing in conductor ‘a’ at any point on conductor ‘b’

$$B_a = \frac{\mu_0 I_a}{2\pi d}$$

(Acting perpendicular downward)

Therefore force on conductor ‘b’ due to field B_a

$$\vec{F} = I_b (\vec{l_b} \times \vec{B_a})$$

$$|\vec{F_{ba}}| = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$$

$$= \frac{\mu_0 I_a I_b l_b}{2\pi d}$$

$$\frac{|\vec{F_{ba}}|}{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$$

Definition of 1 A :

Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7} Nm^{-1}$, when kept 1m apart in vacuum

(b) $M = NI\pi a^2$

$$= 5 \times 2 \times \frac{22}{7} \times 49 \times 10^{-4}$$

$$= 154 \times 10^{-3}$$

$$= 0.154 Am^2$$

$\vec{M}$ will be perpendicular to $x - y$ plane or parallel to Z axis

1/2

1/2

1/2

1/2

1

1/2

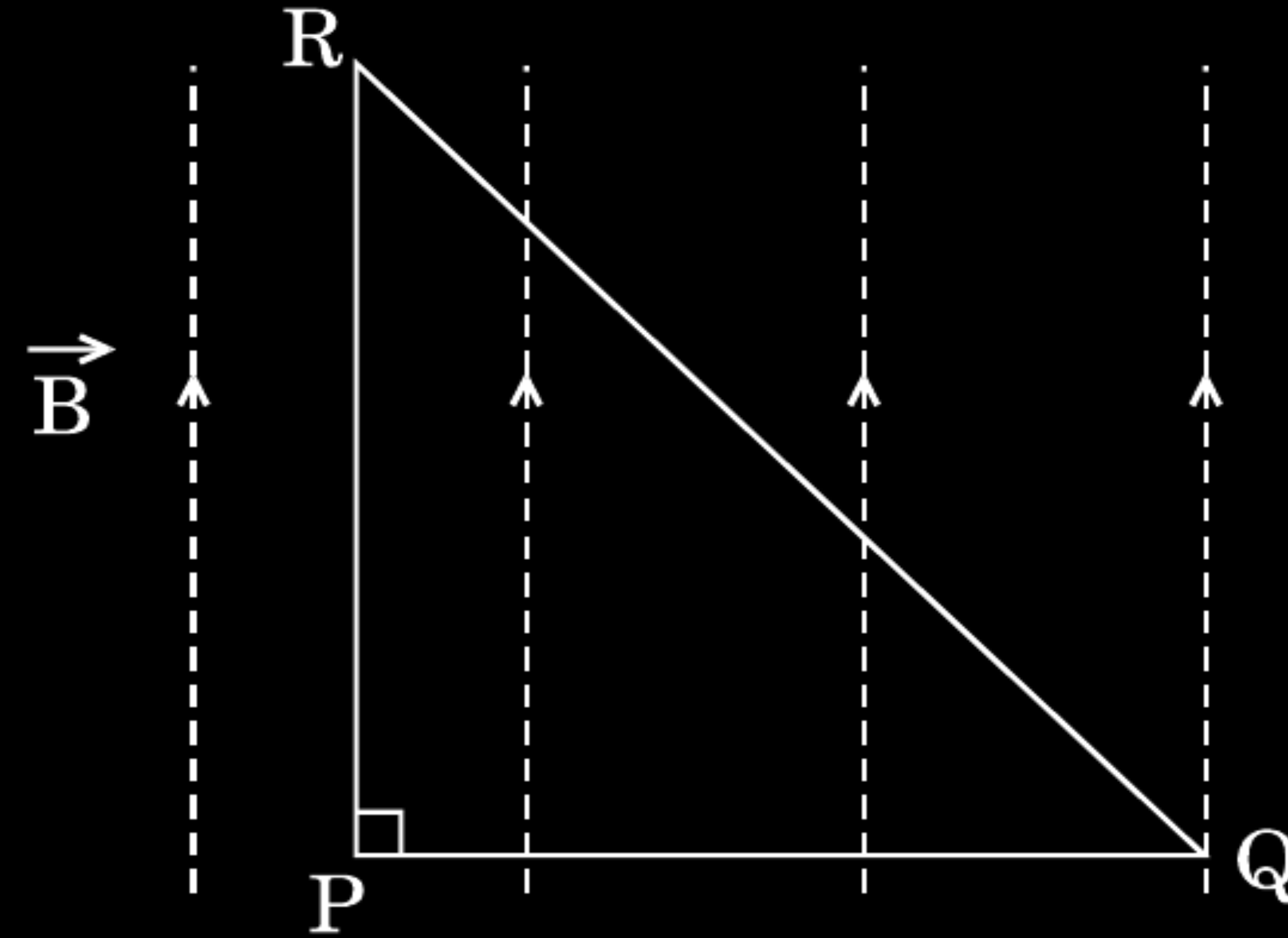
1/2

1/2

1/2

10. An isosceles right angled current carrying loop PQR is placed in a uniform magnetic field $\vec{B}$ pointing along PR. If the magnetic force acting on the arm PQ is F, then the magnetic force which acts on the arm QR will be

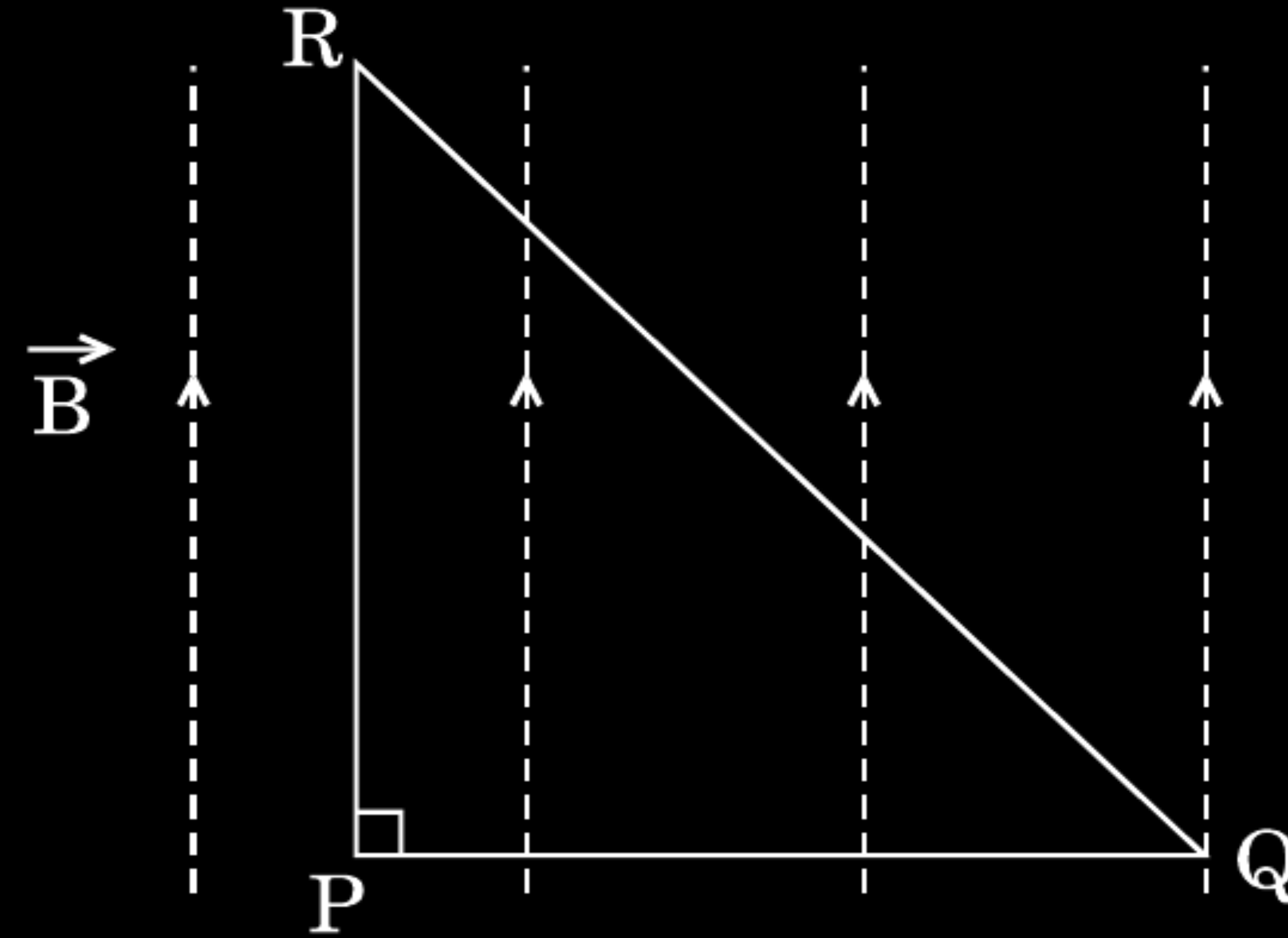
1



- (A) F
- (B) $\frac{F}{\sqrt{2}}$
- (C) $\sqrt{2} F$
- (D) $-F$

10. An isosceles right angled current carrying loop PQR is placed in a uniform magnetic field $\vec{B}$ pointing along PR. If the magnetic force acting on the arm PQ is F, then the magnetic force which acts on the arm QR will be

1



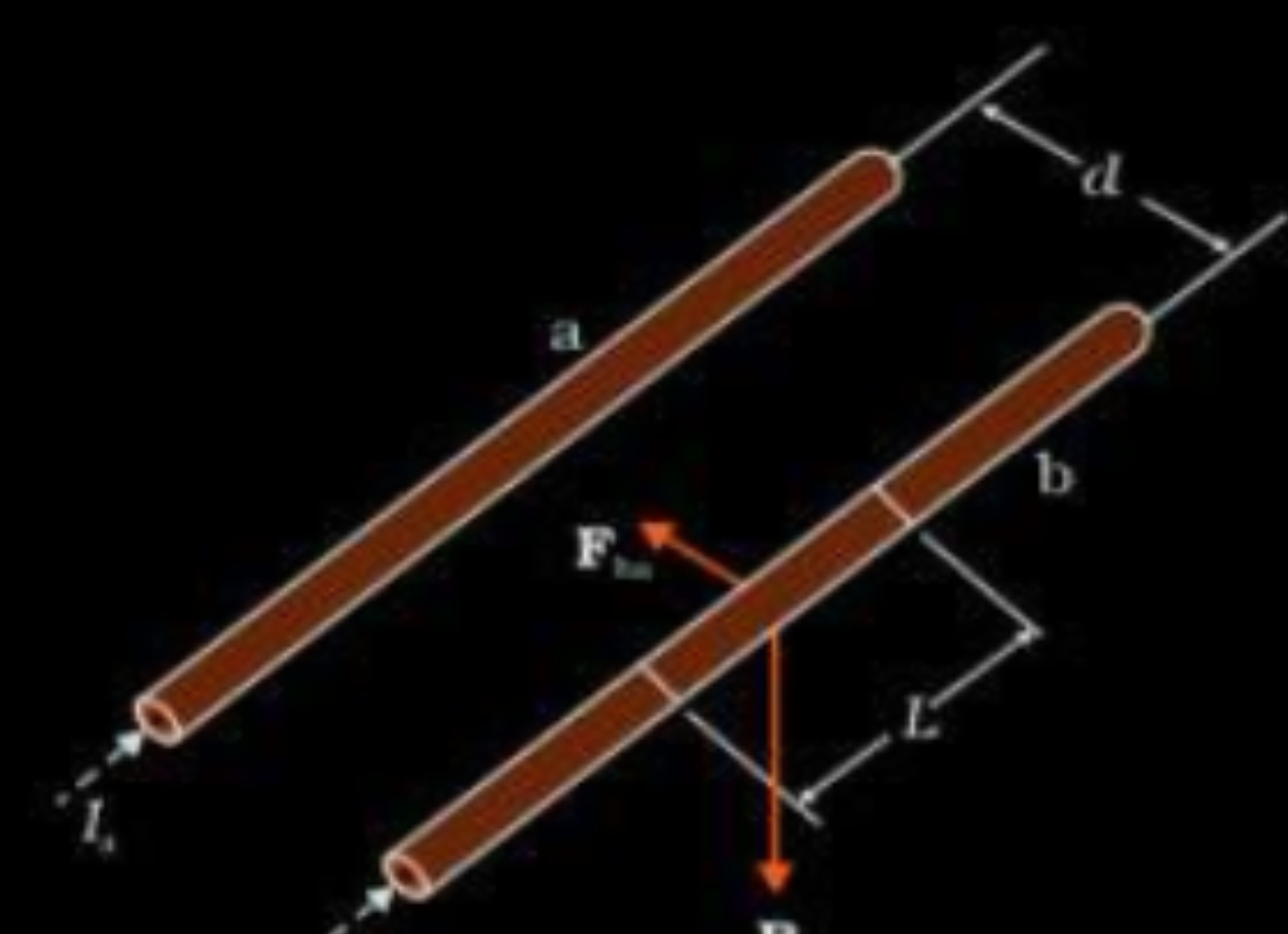
- (A) F
- (B) $\frac{F}{\sqrt{2}}$
- (C) $\sqrt{2} F$
- (D) $-F$

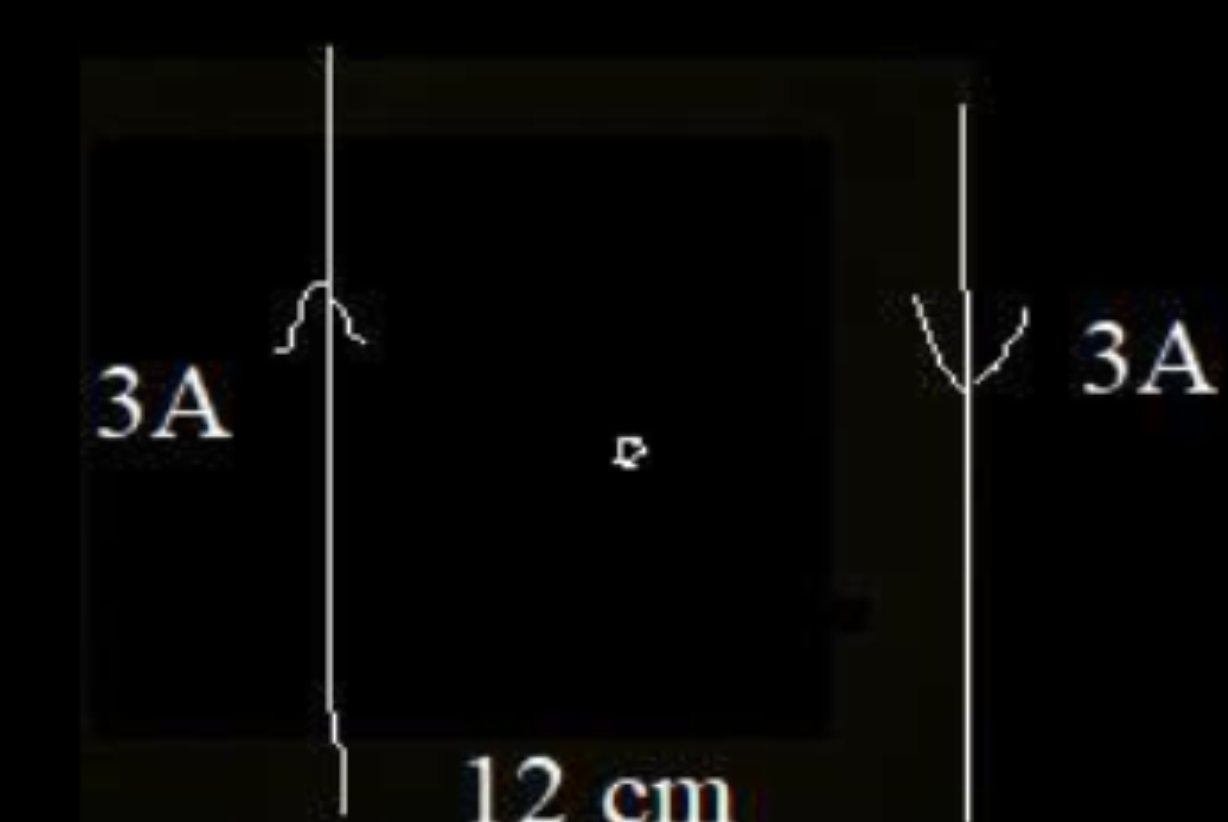
(D)

$-F$

- 36.** (a) Derive the expression for the force acting per unit length between two long straight parallel current carrying conductors. Hence define one ampere.
- (b) Two long parallel straight conductors are placed 12 cm apart in air. They carry equal currents of 3 A each. Find the magnitude and direction of the magnetic field at a point midway between them (drawing a figure) when the currents in them flow in opposite directions.

a) Diagram showing direction	½ mark
Derivation of expression	1 ½ marks
Definition of 1 ampere	1 mark
b) Magnetic fields of the wires	½ mark
Net magnetic field and its direction	1 + ½ mark

(a) 	1/2
Magnetic field due to the current I_a flowing in conductor 'a' at any point on conductor 'b' $B_a = \frac{\mu_0 I_a}{2\pi d}$	1/2
(Acting perpendicular downward) Therefore force on conductor 'b' due to field B_a	1/2
$\vec{F} = I_b (\vec{l}_b \times \vec{B}_a)$ $ \vec{F}_{ba} = I_b l_b \times \frac{\mu_0 I_a}{2\pi d}$ $= \frac{\mu_0 I_a I_b l_b}{2\pi d}$ $\frac{ \vec{F}_{ba} }{l_b} = \frac{\mu_0 I_a I_b}{2\pi d}$	1/2
Definition of 1 A : Two straight infinitely long parallel conductors are said to carry 1 A current each when they interact each other with a force of $2 \times 10^{-7} \text{ Nm}^{-1}$, when kept 1m apart in vacuum	1

	1/2
$\vec{B} = \vec{B}_1 + \vec{B}_2$	
$B = \frac{\mu_0 I_1}{2\pi r_1} + \frac{\mu_0 I_2}{2\pi r_2}$	
$= \frac{\mu_0}{2\pi} \left(\frac{3}{6 \times 10^{-2}} + \frac{3}{6 \times 10^{-2}} \right)$	
$= \frac{4\pi \times 10^{-7} \times 3}{\pi \times 6 \times 10^{-2}}$	
$= 2 \times 10^{-5} \text{ tesla}$	1
Direction of $\vec{B}$ at midpoint is perpendicular to the plane containing the two conductors and pointing downwards. (Note: give full credit of this direction if student takes direction opposite to the shown in fig and answer accordingly)	1/2

- 36.** (a) A circular loop of radius R carries a current I . Obtain an expression for the magnetic field at a point on its axis at a distance x from its centre.
- (b) A conducting rod of length 2 m is placed on a horizontal table in north-south direction. It carries a current of 5 A from south to north. Find the direction and magnitude of the magnetic force acting on the rod. Given that the Earth's magnetic field at the place is 0.6×10^{-4} T and angle of dip is $\frac{\pi}{6}$.

5

(a) Magnetic field at a point on the axis of the current loop	3
(b) Magnitude and direction of the magnetic force	2

$$dB = \frac{\mu_0 Idl}{4\pi(x^2 + R^2)}$$

dB has two components dB_x and $dB_{\perp}$, perpendicular components from diametrically opposite elements dl cancel out, thus only dB_x components remain effective

$$dB_x = dB \cos\theta$$

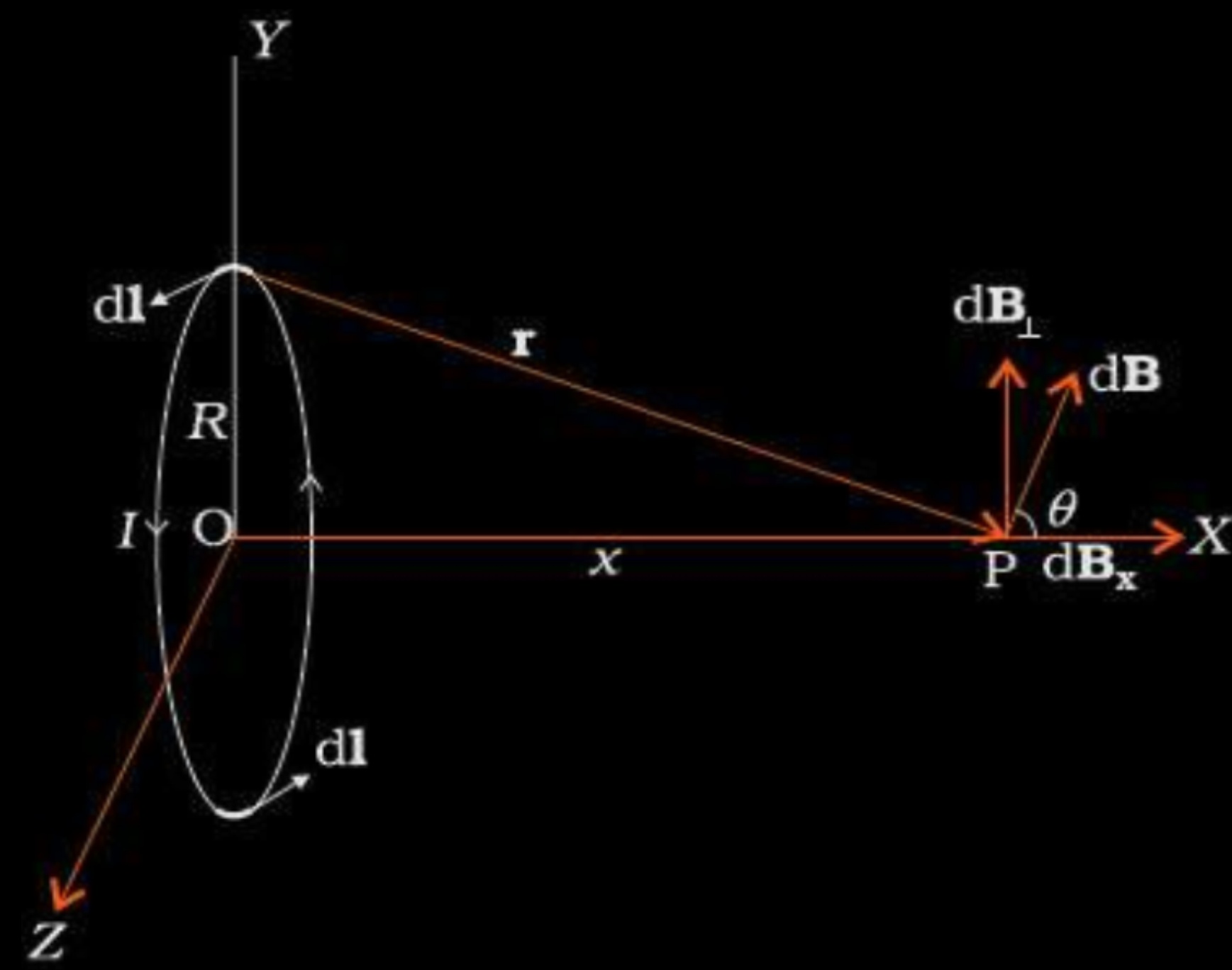
$$\text{and } \cos\theta = \frac{R}{(x^2 + R^2)^{\frac{1}{2}}}$$

$$\therefore B = \int dB_x$$

$$= \int_0^{2\pi R} \frac{\mu_0 IdlR}{4\pi(x^2 + R^2)^{\frac{3}{2}}}$$

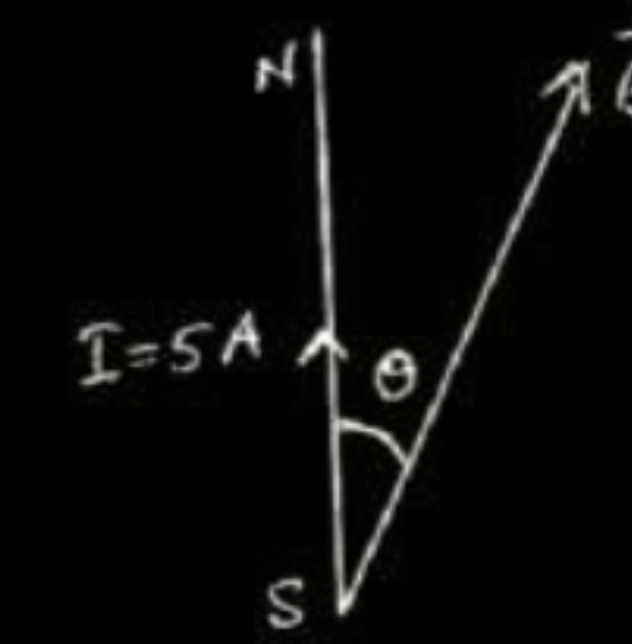
$$= \frac{\mu_0 IR^2}{2(x^2 + R^2)^{\frac{3}{2}}}$$

along the axis of the loop



(b)

(i)



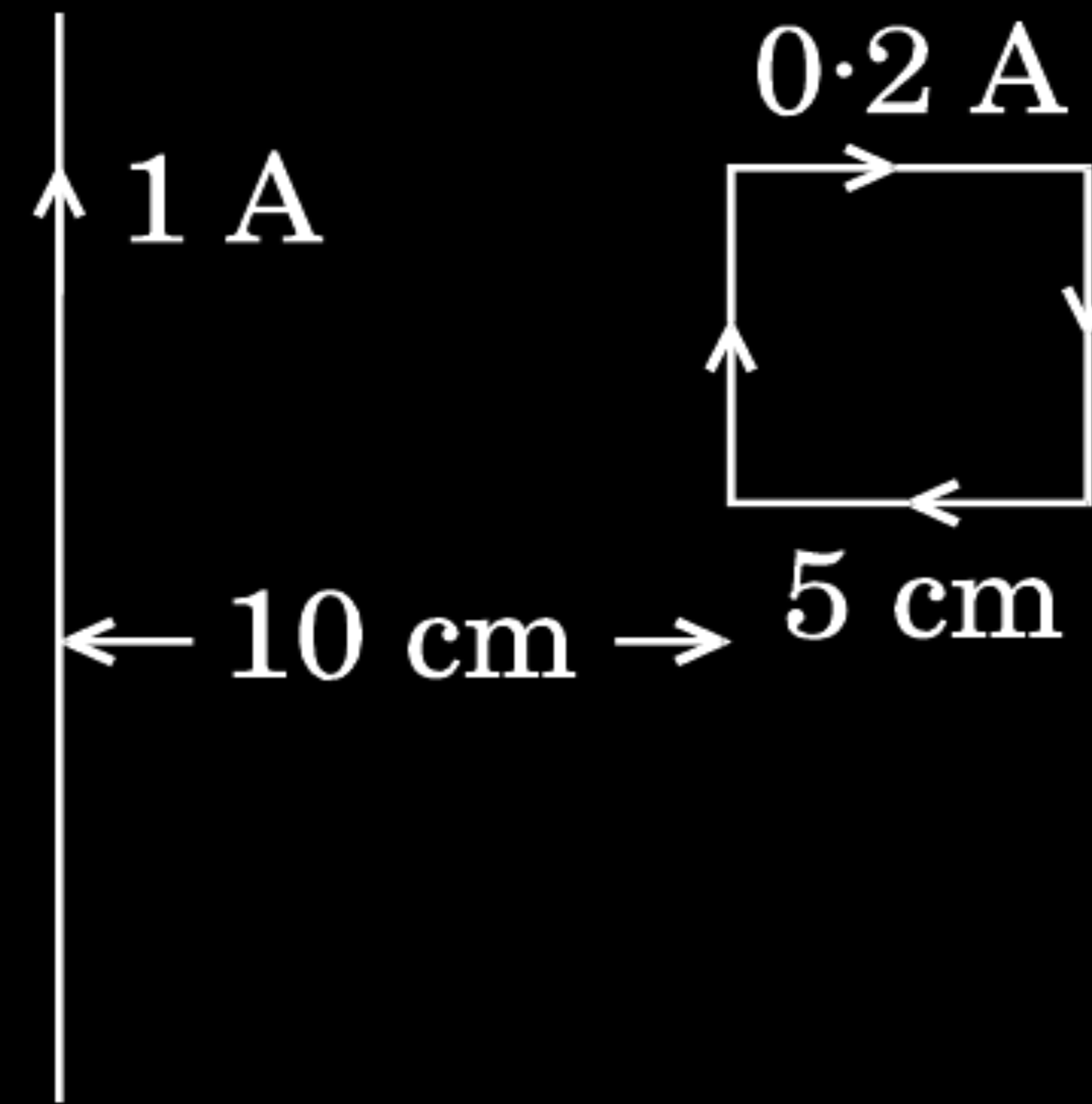
$$\vec{F} = I[\vec{l} \times \vec{B}]$$

$$F = IlB \sin\theta$$

$$= 5 \times 2 \times 0.6 \times 10^{-4} \times 0.5 = 3 \times 10^{-4} \text{ N}$$

(ii) Towards east

- (b) A square loop of sides 5 cm carrying a current of 0.2 A in the clockwise direction is placed at a distance of 10 cm from an infinitely long wire carrying a current of 1 A as shown. Calculate (i) the resultant magnetic force, and (ii) the torque, if any, acting on the loop.



Force of attraction experienced by the length SP of the loop per unit length

$$F_1 = \frac{2\mu_0 I_1 I_2}{4\pi r_1}$$

$$f_1 = \frac{2 \times 10^{-7} \times 1 \times 0.2}{10 \times 10^{-2}} = 4 \times 10^{-7} \text{ Nm}^{-1}$$

Force is attractive

$$f_2 = \frac{2\mu_0 I_1 I_2}{4\pi r_2}$$

$$f_2 = \frac{2 \times 10^{-7} \times 1 \times 0.2}{15 \times 10^{-2}} = 2.6 \times 10^{-7} \text{ Nm}^{-1}$$

Force is repulsive

So the net force experienced by the loop is (per unit length)

$$f = (f_1 - f_2)$$

Total force experienced by the loop is:

$$F = (f_1 - f_2)l = (1.4 \times 10^{-7}) \times 5 \times 10^{-2}$$

$$F = 7 \times 10^{-7} \text{ N}$$

Net force is attractive in nature

As the lines of action of forces coincide torque is zero.

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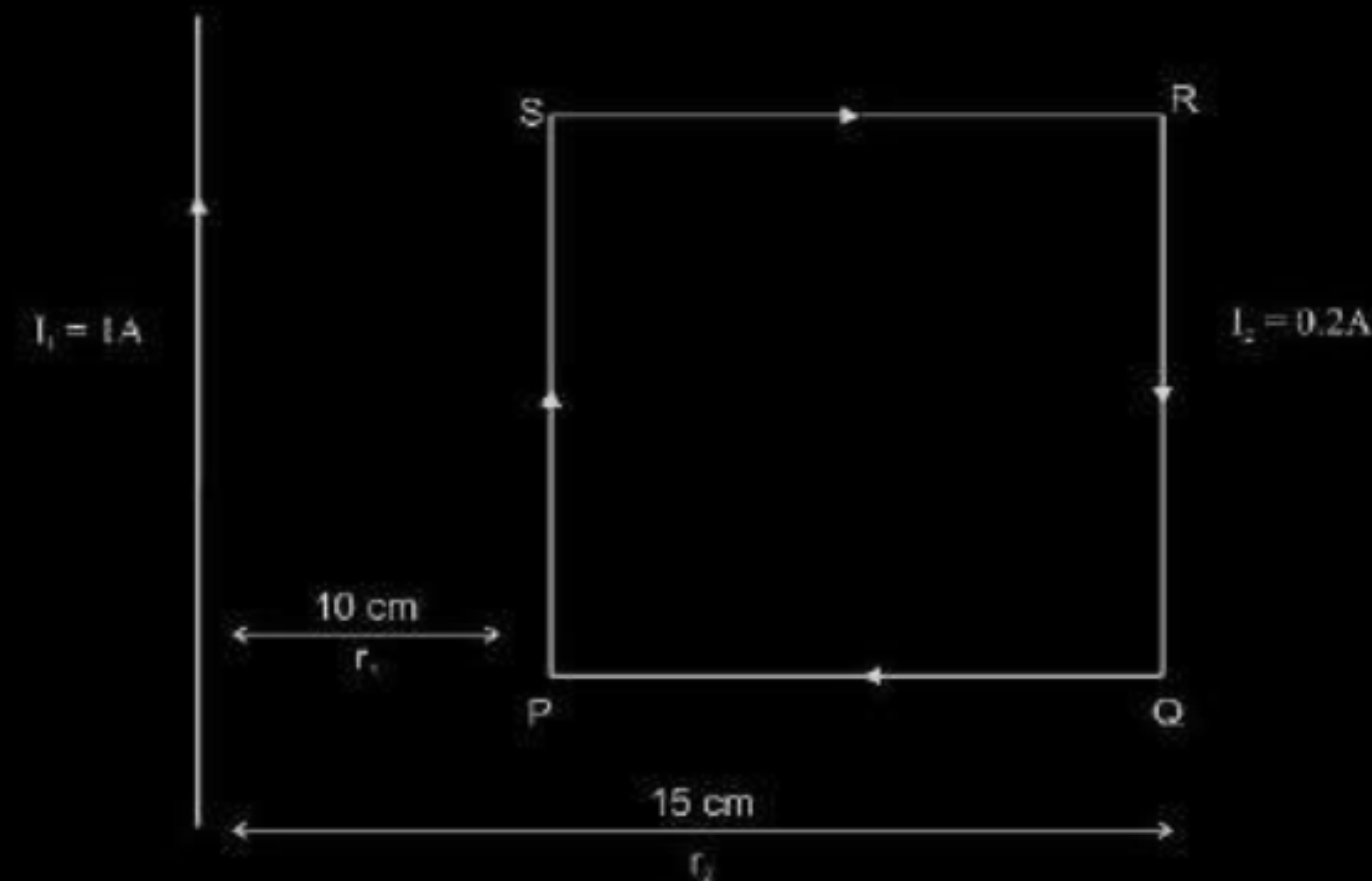
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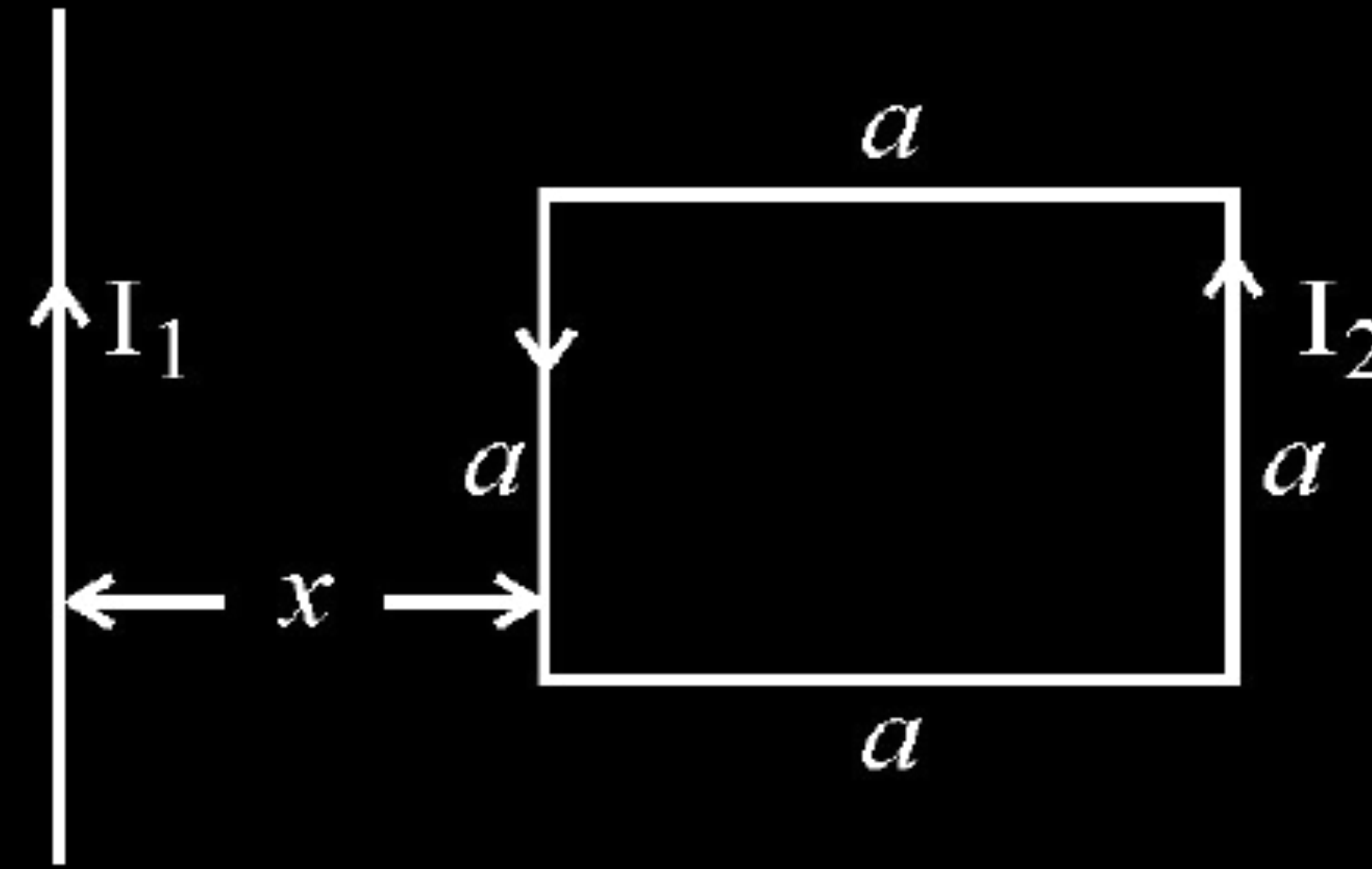
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- (b) A square loop of side 'a' carrying a current I_2 is kept at distance x from an infinitely long straight wire carrying a current I_1 as shown in the figure. Obtain the expression for the resultant force acting on the loop.



(b) Force per unit length between two parallel straight conductors

$$F = \frac{\mu_0}{4\pi} \frac{2I_1 I_2}{d}$$

Force on the part of the loop which is parallel to infinite straight wire and at a distance x from it.

$$F_1 = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a}{x} \quad (\text{away from the infinite straight wire})$$

Force on the part of the loop which is at a distance $(x + a)$ from it

$$F_2 = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a}{(x + a)} \quad (\text{towards the infinite straight wire})$$

Net force $F = F_1 - F_2$

$$F = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a}{x(x + a)} \left[\frac{1}{x} - \frac{1}{x + a} \right]$$

$$F = \frac{\mu_0}{2\pi} \frac{I_1 I_2 a^2}{x(x + a)} \quad (\text{away from the infinite straight wire})$$

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- (a) Define SI unit of current in terms of the force between two parallel current carrying conductors.
- (b) Two long straight parallel conductors carrying steady currents I_a and I_b along the same direction are separated by a distance d . How does one explain the force of attraction between them ? If a third conductor carrying a current I_c in the opposite direction is placed just in the middle of these conductors, find the resultant force acting on the third conductor.

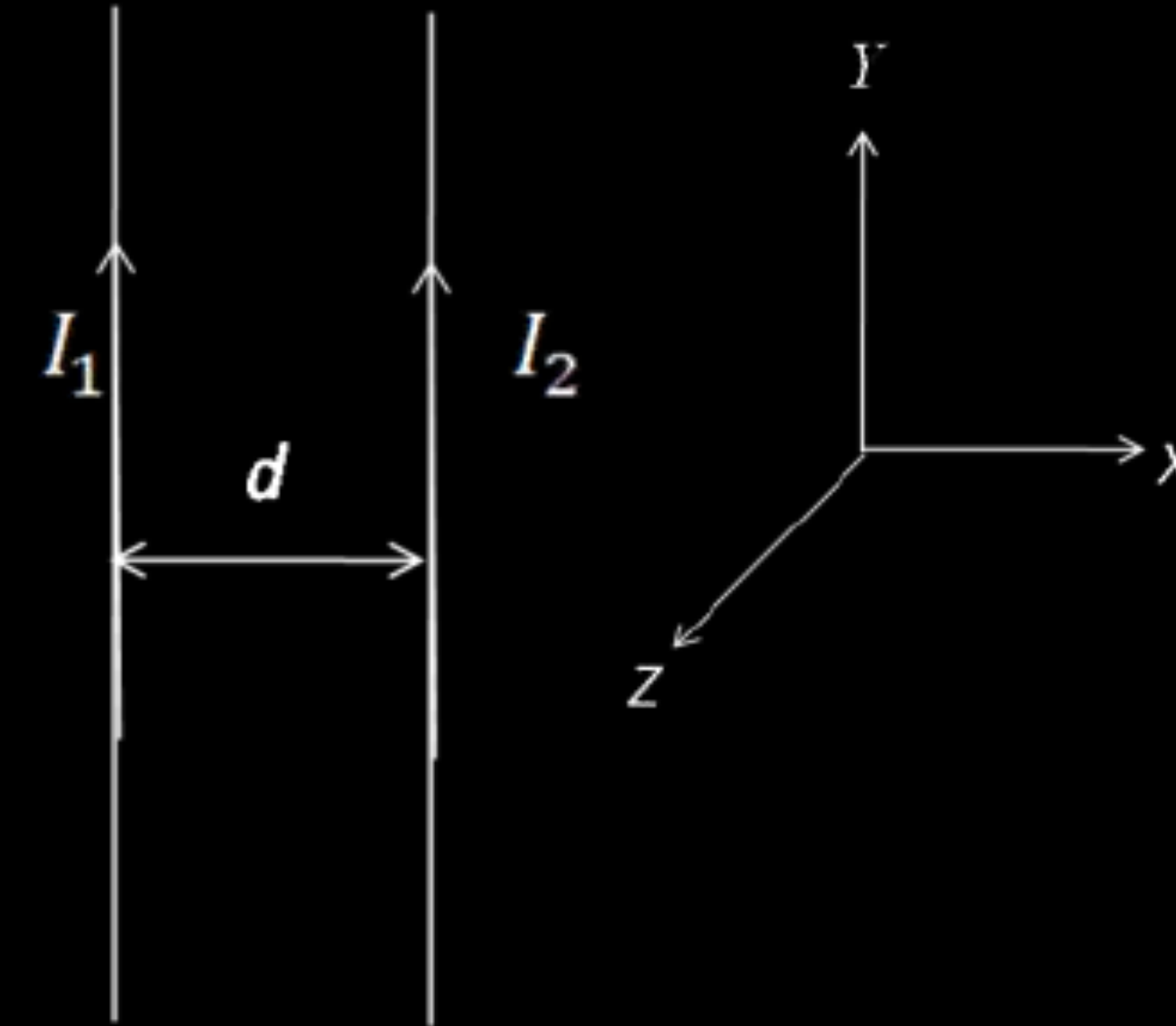
Two long straight parallel conductors carry steady current I_1 and I_2 separated by a distance d . If the currents are flowing in the same direction, show how the magnetic field set up in one produces an attractive force on the other. Obtain the expression for this force. Hence define one ampere.

Diagram showing attractive force on other wire.	1
Obtaining an expression for force.	1
Definition of one ampere.	1

As shown in Figure, the direction of force on conductor b is attractive
[Alternatively:

$\vec{B}$ at a point on wire 2, is along $-\hat{k}$

$\therefore \vec{F}$, on wire 2, due to the $\vec{B}$, is along $-\hat{i}$, i.e. towards wire1. Hence the force is attractive.



Magnetic field, due to current in conductor a,

$$B_1 = \frac{\mu_0 I_1}{2\pi d}$$

The magnitude of force on a length L of conductor b,

$$F_2 = I_2 L B_1$$

$$F_2 = \frac{\mu_0 I_1 I_2 L}{2\pi d}$$

One ampere is that steady current which, when maintained in each of the two very long, straight, parallel conductors, placed one meter apart in vacuum, would produce on each of these conductors a force equal to 2×10^{-7} newton per meter of their length.

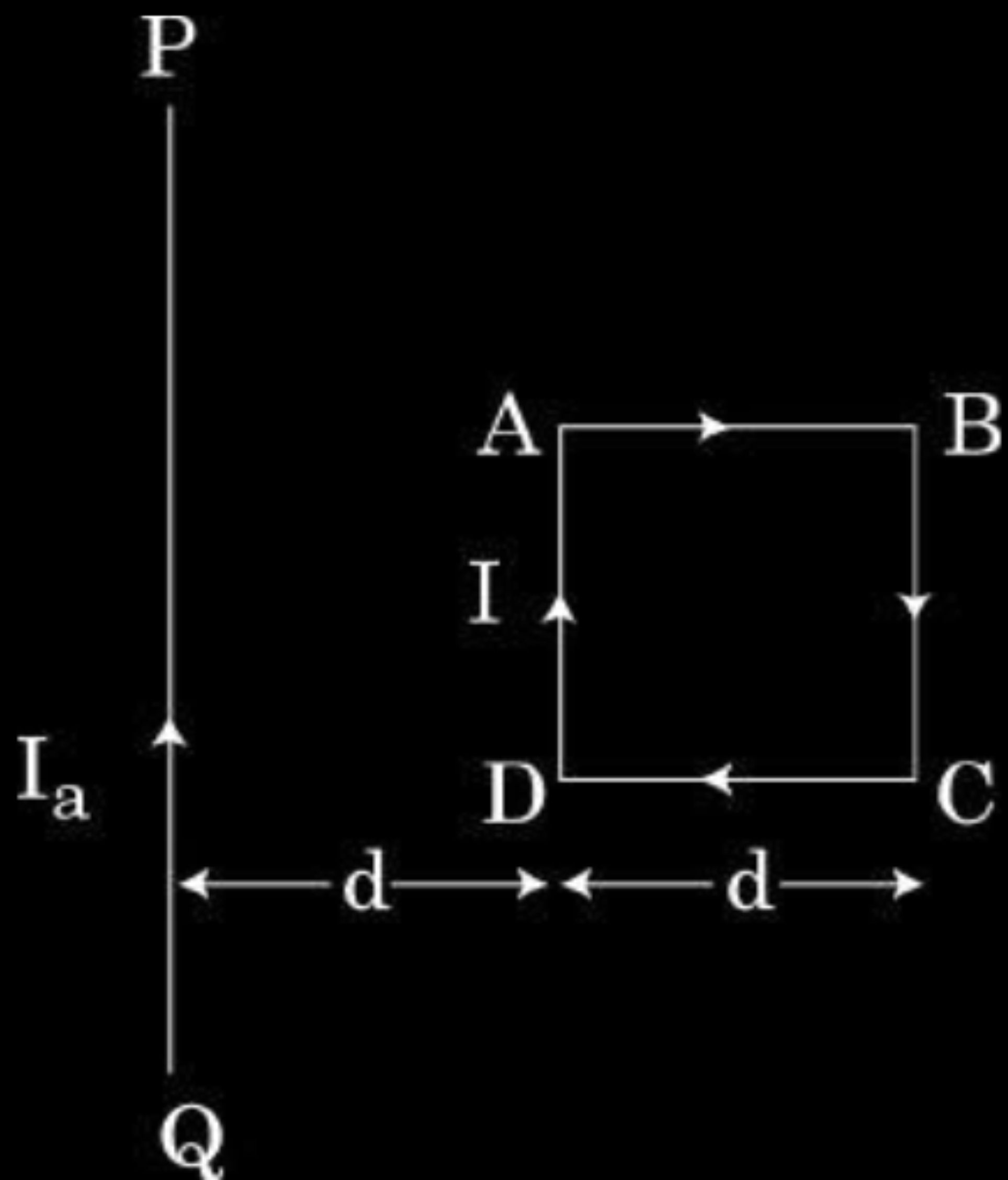
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- (i) Depict magnetic field lines due to two straight, long, parallel conductors carrying steady currents I_1 and I_2 in the same direction.
- (ii) Write the expression for the magnetic field produced by one of the conductor over the other. Deduce an expression for the force per unit length.
- (iii) Determine the direction of this force.
- (iv) In figure given below, wire PQ is fixed while the square loop ABCD is free to move under the influence of currents flowing in them. State with reason, in which direction does the loop begin to move or rotate ?



(ii) $B_1 = \frac{\mu_0 I_1}{2\pi d}$ or $\frac{\mu_0 I_2}{2\pi d} = B_2$

$$F = F_{12} = F_{21} = I_1 B_2 L = I_2 B_1 L$$

$$= \left(\frac{\mu_0 I_1 I_2}{2\pi d} \right) L$$

Force per unit length

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

(iii) Attractive force

(iv) Loop ABCD will move towards wire PQ.

Current in wire PQ and Current in arm AD are in the same direction, so they attract each other.

Current in wire PQ and Current in arm BC are in opposite direction, so they repel each other.

Contribution due to current in AB and CD nullify each other.

Since arm AD is nearer than arm BC to arm PQ, so net force on the loop is attractive. Therefore, the loop will move towards the wire PQ.

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